Available online at www.sciencedirect.com
ScienceDirect
Precambrian Research 150 (2006) 221–256
Precambrian Research
www.elsevier.com/locate/precamres

Paleoproterozoic (2155–1970 Ma) evolution of the Guiana Shield (Transamazonian event) in the light of new paleomagnetic data from French Guiana

H. Théveniaut [a,*], C. Delor [a], J.M. Lafon [b], P. Monié [c], P. Rossi [a], D. Lahondère [a]

[a] *BRGM, CDG/CG, 3 Avenue C. Guillemin, B.P. 6009, 45060 Orléans Cedex, France*
[b] *Centro de Geociências, Cx. Postal 8608, Universidade Federal do Pará, 66075-110 Belém, PA, Brazil*
[c] *UMR 5573 Laboratoire Dynamique de la Lithosphère, Université Montpellier 2, 34095 Montpellier Cédex 05, France*

Received 13 December 2005; received in revised form 13 July 2006; accepted 2 August 2006

Abstract

We present a comprehensive paleomagnetic study on Paleoproterozoic (2173–2060 Ma) plutonic and metamorphic rocks from French Guiana, representative of the full range of the main Transamazonian tectonothermal steps. Twenty-seven groups of directions and poles were obtained from combination of 102 sites (613 samples) based on age constraint, similar lithology and/or geographical proximity. Paleomagnetic results show variations between rocks of different ages which are supposed to be characteristic of magnetizations acquired during uplift and cooling of successive plutonic pulses and metamorphic phases. This is also reinforced by positive field tests (baked contact and reversal tests). Recent U/Pb and Pb/Pb on zircon and complementary $^{40}Ar/^{39}Ar$ on amphibole and biotite allow questioning the problem of magnetic ages relative to rock formation ages. Estimated magnetic ages, based on amphibole dating as a proxy, enable us to construct a Guiana Shield apparent polar wander path for the 2155–1970 Ma period. It is also possible to present paleolatidudinal evolution and continental drift rates related to specific Transamazonian tectonic regimes.

French Guiana and probably the Guiana Shield were located at the Equator from *ca*. 2155 to 2130 Ma during the Meso-Rhyacian D1 magmatic accretion phase, related to subduction of Eorhyacian oceanic crust. After closure of the Eorhyacian Ocean and collision of West African and Amazonian plates, the Guiana Shield moved. The first evolution towards 60° latitude, occurs after 2080 Ma, during the Neorhyacian D2a post collisional sinistral transcurrent phase. During the Late Rhyacian D2b phase, up to 2050 Ma, the Guiana Shield reaches the pole and starts to move to lower latitudes on an opposite meridian. By the Orosirian D2c phase, from *ca.* 2050 to 1970 Ma, the Guiana Shield reaches the Equator.

Based on the amphibole $^{40}Ar/^{39}Ar$ dates, we estimate the continental drift between 12 and 16 cm/y for the Meso to Late Rhyacian period followed by a lower rate between 9 and 14 cm/y up to Orosirian time. This study highlights rock ages and magnetic ages are prerequisite to any continental reconstruction especially when it is shown continental drift is important for a 100–200 Ma time period. Our results confirm the possibility of APWP construction on Paleoproterozoic plutonic rocks but suggest improvement will rely on the combination with multidisciplinary approaches such as structural geology and multi-method radiometric dating.
© 2006 Elsevier B.V. All rights reserved.

Keywords: Paleomagnetism; Paleoproterozoic; Transamazonian; French Guiana; Continental drift

1. Introduction

In the late 1950s, pioneering paleomagnetic work (Runcorn, 1956; Irving, 1956) played a key role in the understanding of Phanerozoic Plate Tectonics. It was

* Corresponding author. Tel.: +33 2 38 64 34 55; fax: +33 2 38 64 36 85.
E-mail address: h.theveniaut@brgm.fr (H. Théveniaut).

0301-9268/$ – see front matter © 2006 Elsevier B.V. All rights reserved.
doi:10.1016/j.precamres.2006.08.004

well demonstrated that apparent polar wander paths from various continents were similar but differed in a systematic way which could only be resolved by the progressive opening of the oceans. This led to the demonstration that Phanerozoic orogenic belts were mainly due to continental drift and successive opening and closure of oceans. Since then, most of paleomagnetic studies have been concentrated on Phanerozoic rocks to ascertain or better constrain first, second or sometimes third order configuration of the continents at different time periods.

Precambrian syntheses of paleomagnetic data (McElhinny and Mc Williams, 1977; Piper, 1982, 1987) led to the opposite conclusion that a Precambrian supercontinent probably survived for most of the Proterozoic time period as no significant displacement was associated with the successive orogenic belts. They concluded that Plate Tectonics was not the dominant tectonic mechanism for Precambrian time to account for these orogenic belts. However, the synthesis of Van der Voo and Meert (1991) on Neoproterozoic to Cambrian paleomagnetic data brought a new critical appraisal of the possible tectonic models based on paleomagnetic data. They concluded that no clear paleomagnetic arguments favoured any of the dualistic models to account for orogenic belts. These models are the ensialic one which favours no relative motion between continents while the ensimatic model corresponds to the Plate Tectonic model accepted for Phanerozoic time. Van der Voo and Meert (1991) showed, on the basis of selected criteria used for Phanerozoic paleomagnetic data, both models could be acceptable but any tectonic interpretation should be premature for the Neoproterozoic time period. Indeed, the arbitrary choice of paleomagnetic polarity, the low precision of pole dating and the scarcity of reliable paleomagnetic results (number of sites, of samples, proper demagnetizations and field tests) precluded any hasty conclusion. If this is totally acceptable for Neoproterozoic time it can stand for earlier periods such as the Paleoproterozoic. However, as new well dated paleomagnetic data arise, Neoproterozoic supercontinent proposal becomes possible (Weil et al., 1998). Recent review of global 2.1–1.8 Ga orogens (Zhao et al., 2002) confirm Van der Voo and Meert (1991) approach for a Paleoproterozoic supercontinent. Geological arguments favour a possible major continental assembly with existence of the Columbia Supercontinent (Rogers and Santosh, 2002) during the 2.1–1.8 Ga time period. However, paleomagnetic data are still too scarce to validate any proposed configuration. A recent synthesis (Pesonen et al., 2003) proposed different configurations of continents during the Proterozoic. The 2.0 Ga Paleoproterozoic reconstruction of the "Kenorland" supercontinent is based on data with similar ages but derived from different method. Even though tentative and geologically acceptable, such reconstruction may take into account the possible difference, due to the cooling rate, between U/Pb and Ar/Ar ages. Indeed, a difference of more than 100 Ma may exist on a same rock with these two methods as discussed hereafter. It is therefore difficult to compare data which may differ significantly in age for Paleoproterozoic periods which no one could do for Mesozoic period.

Proterozoic paleomagnetic studies of the Guiana Shield were initially undertaken on Proterozoic doleritic dikes from Venezuela and Guyana (Hargraves, 1968, 1978), and in Suriname (Veldkamp et al., 1971). This was followed by studies on the different stratigraphic units in Venezuela (Onstott and Hargraves, 1981; Onstott et al., 1984) and more recently by studies of Mesoproterozoic sedimentary (Castillo and Constanzo-Alvarez, 1993) and plutonic (Miron-Valdespino and Constanzo-Alvarez, 1997) units in Venezuela. The Paleoproterozoic results (Onstott and Hargraves, 1981; Onstott et al., 1984) were compared with West African ones to derive a relative motion model between the South American and African Shields during the Paleoproterozoic. Although some of these results are questionable (see Section 5), this was, at the onset of the new millennium, the only Paleoproterozoic paleomagnetic basis for the Guiana Shield.

Within the framework of the multidisciplinary 1:500,000 geological updating mapping program carried out by BRGM in French Guiana Territory (Delor et al., 2001, 2003a), we had the opportunity to sample most of outcropping rock formations. Apart from Early Jurassic and possibly scarce Proterozoic doleritic dikes, all plutonic and volcanosedimentary formations belong to the Rhyacian era (2300–2050 Ma) and coeval Transamazonian tectono-thermal event (2.26–1.95 Ga) which affected the Guiana Shield with no evidence of later event which could have induced any possible remagnetization. The paleomagnetic sampling (1995–2001) in French Guiana was the first opportunity to investigate the paleomagnetic signal of all representative plutonic and metamorphic rock units of the territory. The aim of this work was therefore to obtain Paleoproterozoic paleomagnetic directions and well calibrated poles in order to derive an Apparent Polar Wander Path (APWP) for the Guiana Shield as a basis for the understanding of the geodynamic history at the time of emplacement (uplift and cooling) of the different formations. Jurassic doleritic dikes were also sampled during the successive field trips. The results (Nomade et al., 2000) will only be discussed hereafter when the possibility of a baked contact test

was performed. Recently, Nomade et al. (2001, 2003) presented preliminary results of these comprehensive paleomagnetic sampling along the Oyapock, the Maroni and the Mana rivers. Three paleomagnetic poles were derived and dated using $^{40}Ar/^{39}Ar$ cooling ages. They were also compared to new paleomagnetic results from West Africa (Nomade et al., 2003). In French Guiana, these results were obtained and discussed prior to the publication of the new French Guiana geological map (Delor et al., 2001) better constrained by compilation of field observations, aero-geophysical data and acquisition of about 100 new isotopic data (Pb–Pb, U–Pb, U–Th–Pb and Sm–Nd). In this paper, some results of Nomade et al. (2003) will be reinterpreted, in Section 5, in the light of the recent geological overview (Delor et al., 2003a) of French Guiana Paleoproterozoic terrains, and the subsequent reassessment of Paleoproterozoic formations in terms of absolute ages referring to Eo-, Meso- and Neo to Late Rhyacian ages.

2. Geological setting

2.1. French Guiana and the Guiana Shield

French Guiana belongs to the Guiana Shield (Gibbs and Barron, 1993; Delor et al., 2003a,b) which extends from Northwest to Southeast from eastern Venezuela to northern Brazil (Fig. 1a). It is mainly composed by Paleoproterozoic granite-greenstone belts generated during the main Transamazonian tectono-thermal event (2.26–1.95 Ga). Some Archean relicts have been observed in Venezuela (Imataca complex) and in the Amapá and Pará states in Brazil (Lima et al., 1982; João and Marinho, 1982; Montalvão and Tassinari, 1984; Avelar et al., 2003; Rosa-Costa et al., 2003). A Paleoproterozoic volcanic event (Uatumã Supergroup, 2.05–1.8 Ga) followed the main Transamazonian tectono-thermal event and a thick widespread sedimentary cover (Roraima Supergroup) was subsequently deposited on top of it, as shown by ages at *ca.* 1.79 Ga obtained on interlayered dolerite sills (Santos et al., 2002). A few "isolated" magmatic or metamorphic events followed during the Mesoproterozoic and Neoproterozoic up to the deposition of scarce Phanerozoic sedimentary series. The latest recognized event is the extensive doleritic dike swarm emplacement associated with the opening of the Atlantic Ocean in Early Jurassic time.

French Guiana lithologies (Delor et al., 2003a) do not contain relics of Archean basement, apart from inherited minerals in Paleoproterozoic sediments and granites (Avelar et al., 2003). Earliest crustal growth corresponds to the formation of an Eorhyacian juvenile oceanic crust. Relics of this initial stage (2.26–2.20 Ga) have been found in gabbroic rocks in the Atlantic coastal exposures of the Cayenne area and more widely throughout French Guiana, on inherited zircon ages from younger metasedimentary and reworked plutonic rocks.

Between 2.18 and 2.13 Ga, a dominant tonalite-trondhjemite-granodiorite (TTG) magmatism associated with volcano-sedimentary deposits and metamorphism marks the closure of this Paleoproterozoic Ocean. This Mesorhyacian D1 magmatic accretion event has been argued as a multipulse island-arc plutono-volcanism in a subduction context due to the West African and northeast South American Archean blocks convergence. Inside this D1 broad 2.18–2.13 Ga magmatic cycle, a possible diachronism is suggested by the presence of younger TTG magmatism in Central French Guiana (Central TTG Complex at 2.15–2.13 Ga) and older TTG pluton and greenstone belts which border it to the north and to the south (2.18–2.16 Ga).

A second stage of tectonic accretion (D2) marks the closure of the island arc basins and subduction system. Two tectonothermal substages have been individualised: a NeoRhyacian D2a substage, around 2.11–2.08 Ga, is characterised by dominant granitic magmatism emplaced during sinistral wrenching. In northern French Guiana, this phase is associated with the opening of late detrital (pull apart) basins north of the Central TTG Complex. A Late Rhyacian D2b event, dated at 2.07–2.06 Ga is marked by late metaluminous monzogranite emplacement dissecting the detrital basins along WNW–ESE dextral strike-slip corridors.

Following these Neo- to Late Rhyacian events, only dikes and dike swarm intrusions disturbed the Transamazonian belt. A few dikes have been dated at 1.8 and 0.8 Ga while the dominantly N–S oriented dikes, Early Jurassic in age (Deckart et al., 1997; Nomade et al., 2000) are associated to the initial stages of the opening of the Atlantic Ocean. Phanerozoic sedimentary cover is restricted to coastal Quaternary deposits.

Apart from the coastal outcrops in the vicinity of Cayenne and along the coastal road, the geology of the French Guiana territory (Fig. 1b) can only be investigated by canoeing field trips along the main collectors where the intense and deep weathering is cleaned by the rivers and facilitate proper geological investigations. The geological formations which were sampled for paleomagnetic study during the geological mapping program of the French Guiana territory (Delor et al., 2001) are logically presented in terms of rejuvenated age of formations with special references to Eo-, Meso- and Neo to Late Rhyacian stages.

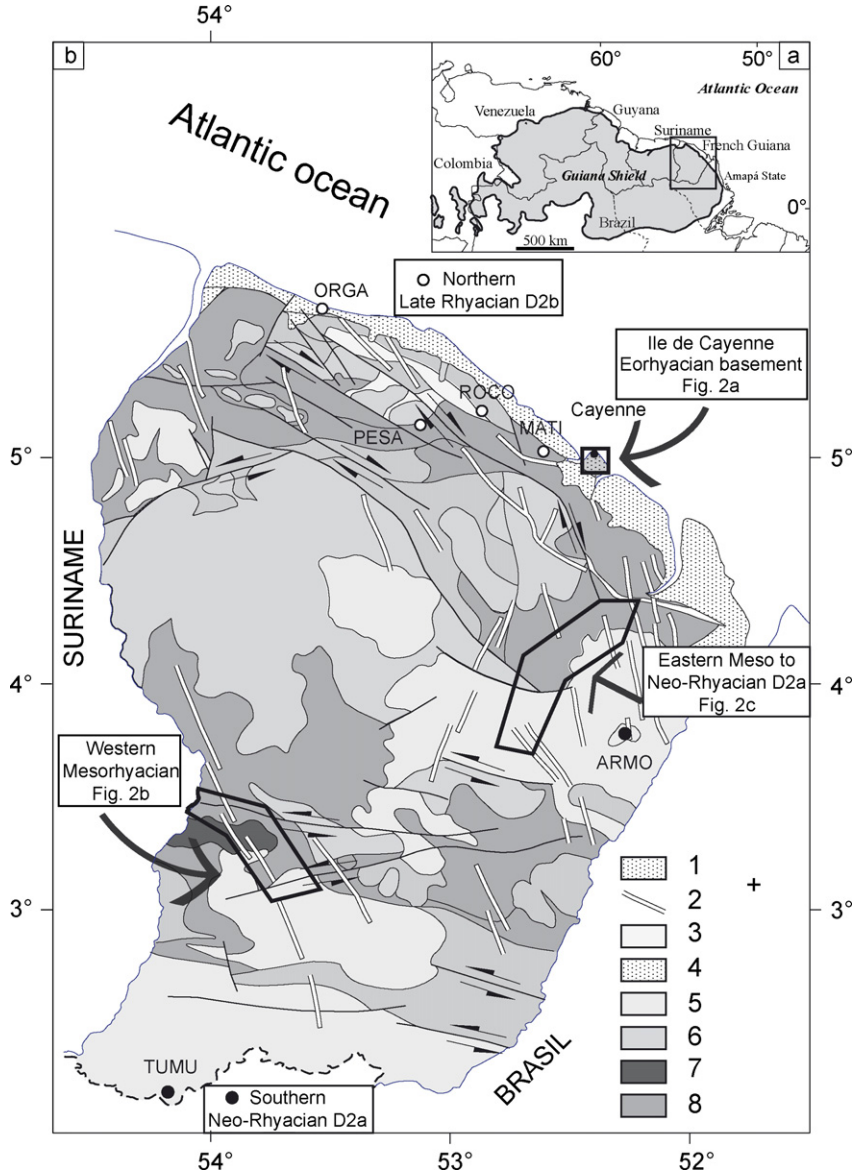


Fig. 1. (a) Northeastern South America and the Guiana Shield in grey (from Delor et al., 2003a). (b) Simplified geological map of French Guiana with the different sampling areas. (1) Meso-Cenozoic cover, (2) Jurassic and Neoproterozoic doleritic volcanism, (3) Late Rhyacian D2a metaluminous monzogranite, (4) Late Rhyacian D2b pull-apart detrital basin, (5) Late Rhyacian D2a granitic plutonism, (6) Mesorhyacian D1 TTG plutonism, (7) Mesorhyacian D1 gabbro and (8) Eo to Mesorhyacian greenstone belt.

2.2. Geology, geochronology and sampling

All ages discussed hereafter (Table 1) are detailed in Delor et al. (2003a) apart from those (Table 2) described in the text. Paleomagnetic sites/groups are mentioned but detailed on the following chapter.

2.2.1. The "Ile de Cayenne" Eorhyacian basement

In the Cayenne area (Fig. 2a), the "Ile de Cayenne" metamorphic Complex and the gabbros and diorites of the Mahury Mountain are well exposed on coastal outcrops (Egal et al., 1994; Vanderhaeghe et al., 1998; Delor et al., 2003a). They are located on the northern edge of the northern sedimentary basins of French Guiana. A trondhjemite sample (CAY18A) gave two U/Pb zircon ages (Vanderhaeghe et al., 1998) of 2217 ± 1.5 and 2173 ± 2.3 Ma which were interpreted as inherited age and crystallisation age, respectively. Paleomagnetic sampling of these formations corresponds to the CAYE01 group. A tonalite sample (CAY12A; Vanderhaeghe et al.,

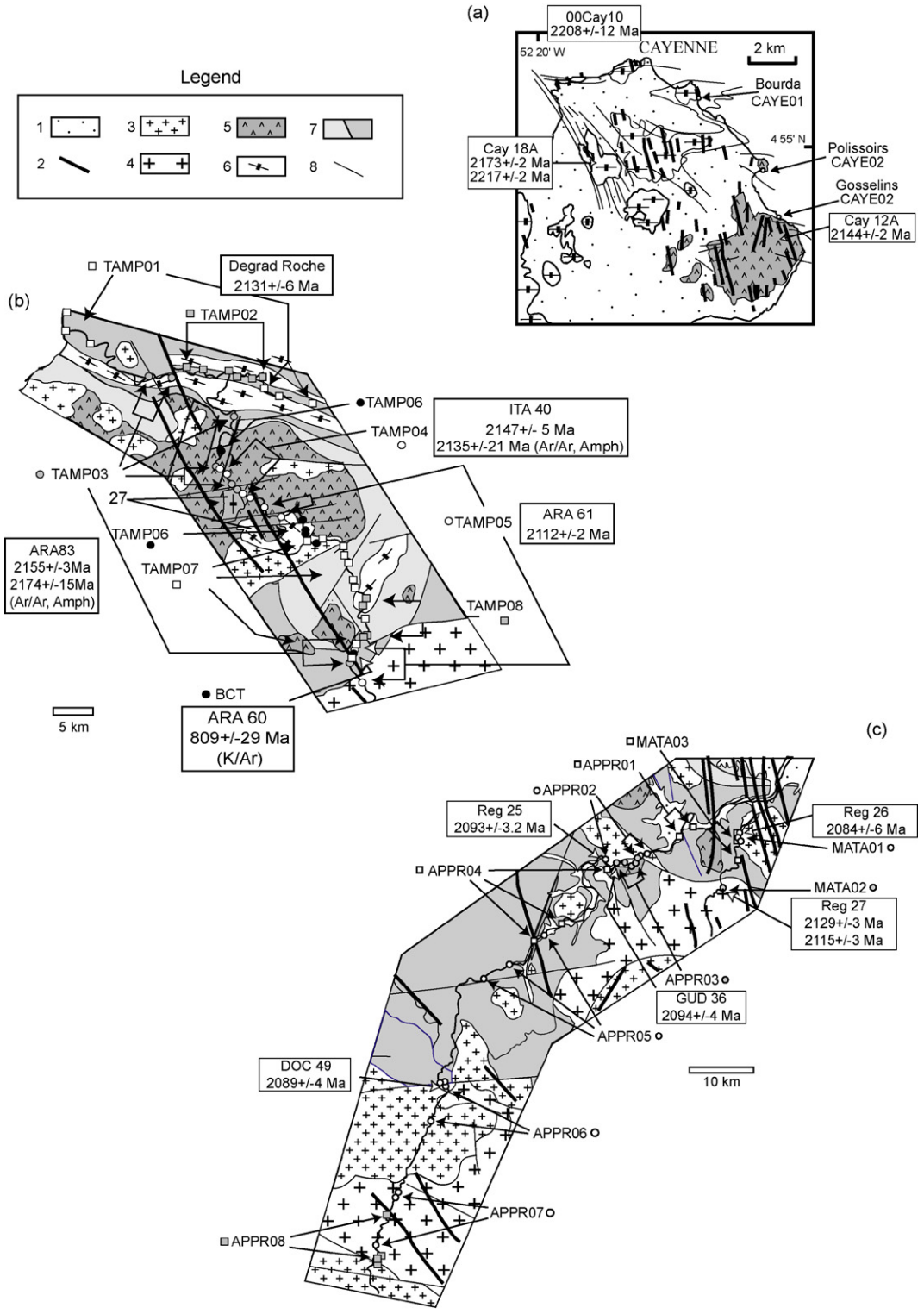


Fig. 2. Simplified geological maps of the Isle de Cayenne EoRhyacian basement (a), the Western Mesorhyacian Tampok/Ouaki rivers area (b), and the Eastern Meso to Late Rhyacian Approuague/Mataroni rivers area (c). (1) Quaternary deposits, (2) Jurassic (a and c) and Neoproterozoic (b) doleritic dikes, (3) Neorhyacian granites, (4) Mesorhyacian TTG intrusions, (5) Meso (a and b) to Neorhyacian (c) gabbros, (6) Mesorhyacian gneisses, (7) Meta-volcano (dark) to sedimentary (clear) formations of the greenstone belt and (8) faults.

Table 1
Synthesis of geochronological results used in this study

Sample identification	Rock type	Age (Z/M) (Ma)	Age (A) (Ma)	Age (B) (Ma)	Paleomagnetic group	Reference
CAY18A	Trondhjemite	2173 ± 2			CAYE01	Vanderhaeghe et al. (1998) recalculated
CAY 12A	Diorite/Tonalite	2144 ± 2			CAYE02	Vanderhaeghe et al. (1998) recalculated
ARA 83A	Granodiorite	2155 ± 3[a]	2174 ± 15	2163 ± 14	TAMP06	Delor et al. (2003a); and this paper with details in Table 2
ITA 40A	Diorite	2147 ± 5	2135 ± 21		TAMP04	Delor et al. (2003a); and this paper with details in Table 2
ARA 61	Granite	2112 ± 2			TAMP05	Delor et al. (2003a)
Degrad Roche	Gneiss	2131 ± 6			TAMP02	Delor et al. (2003a)
REG 27	Tonalite	2115 ± 3			MATA02	Vanderhaeghe et al. (1998) recalculated
TH09	Granite	2100 ± 1		1790 ± 13	TUMU	Vanderhaeghe et al. (1998) recalculated; and this paper with details in Table 2
REG 25	Granite	2093 ± 3			APPR02	Vanderhaeghe et al. (1998) recalculated
DOC 49	Granite	2089 ± 4			APPR06	Delor et al. (2003a)
REG 26	Leucogranite	2084 ± 6			MATA01	Vanderhaeghe et al. (1998) recalculated
GUD36	Gabbro	2094 ± 4			APPR03	Delor et al. (2003a)
ARM001	Mg–K granite	2080 ± 4		1936 ± 12	ARMO	This paper with details in Table 2
Petit Saut	Granite	2060 ± 4			PESA	Delor et al. (2003a)
Organabo	Granite	2069 ± 4			ORGA	Delor et al. (2003a)
89–98RC1	Granite	2095 ± 6[b], 2061 ± 15[b]			ROCO	Delor et al. (2003a)

Age (Z/M): Pb/Pb dating on zircon (TIMS); Age (A): $^{40}Ar/^{39}Ar$ dating on amphibole; Age (B): $^{40}Ar/^{39}Ar$ dating on biotite.
[a] U/Pb dating on zircon (SIMS).
[b] U–Th–Pb dating on monazite.

1998) from the dioritic/tonalitic intrusion of the Mahury Mountain gave a U–Pb zircon age of 2144 ± 1.8 Ma. CAYE02 group belongs to this intrusion.

2.2.2. *The Western Mesorhyacian TTG—greenstone belt (D1 event)*

The Tampok and Ouaki rivers are located in southwestern French Guiana (Fig. 2b) where they follow (Ouaki) or crosscut (Tampok) the southern granite-greenstone belt which extends from West (Suriname) to East (Brazil) on a N120 main direction. The metamorphic rocks mainly consist of gneisses, with rare occurrences of amphibolite and other metagranodiorite while the plutonic rocks are essentially gabbroic to dioritic rocks intruded by a few younger granites and granodiorites. Detailed geological mapping (Hocquard and Schmitt, 1980) and a new geological reappraisal including isotopic data (Delor et al., 2003a)

Table 2
Summary of $^{40}Ar/^{39}Ar$ results obtained during this study

Sample	Total gas date	Plateau date	% ^{39}Ar	Max. age	Min. age	Mean % $^{40}Ar^{*}$
ARA83A amphibole	2160.5 ± 14.3	2173.9 ± 14.6	71	–	–	99.3
ARA83A biotite	2148.1 ± 13.7	2162.9 ± 13.9	85	–	–	99.1
ITA40 amphibole	2111.8 ± 21.0	2134.9 ± 20.7	89	–	–	94.7
ITA40 biotite	–	–	–	2152 ± 14	2016 ± 10	98.4
TH09 amphibole	2026.5 ± 14.4	–	–	–	–	96.5
TH09 biotite	1796.3 ± 12.4	1790.5 ± 12.6	65	–	–	97.8

$^{40}Ar^{*}$ means radiogenic argon.

allow establishing a quite precise chronology of our sampling.

The granodiorite intrusion (ARA83, TAMP06 group), with a U/Pb zircon age (Table 1) of 2155 ± 3 Ma, is the oldest known plutonic intrusion. During this study, $^{40}Ar/^{39}Ar$ laser probe, single grain, step-heating experiments (Maurel et al., 2003) have been carried out (Table 2) on amphibole and biotite from the same sample in order to constrain the cooling history of the pluton down to 300 °C, the blocking temperature for argon in biotite (Harrison et al., 1985). The age spectra of these minerals are displayed in Fig. 3a and b. Amphibole has a convex age pattern and a plateau date of 2174 ± 15 Ma calculated over 70% of the argon released. Its total gas date of 2161 ± 14 Ma reveals that only minor argon loss affected this amphibole. Biotite gives a plateau age of 2163 ± 14 Ma for 85% of the argon released, again with minor evidence of argon loss. The similarity of plateau ages for minerals with closure temperature ranging from 550 °C for amphibole (Dahl, 1996) to 300 °C for biotite, and the agreement with the U/Pb zircon age clearly indicate that the granodiorite experienced fast cooling immediately after its emplacement, with no evidence of later thermal disturbance. Fast cooling probably resulted from the emplacement of the granodiorite in relatively cold upper crustal levels, thus triggering fast heat dissipation.

On geological grounds, the southern orthogneiss (TAMP08 group) seems coeval with the granodiorite (Hocquard and Schmitt, 1980), and locally is truncated by gabbro-dioritic intrusions from the Tampok massif. A mafic intrusion from this massif has been dated in the vicinity of our sampling (ITA40, TAMP04 group). Zircon yields a Pb/Pb age of 2147 ± 5 Ma. Amphibole from this sample provides a flat age spectrum with a plateau date of 2135 ± 21 Ma for 90% of the argon released (Fig. 3c). $^{40}Ar/^{39}Ar$ spot heating experiments have been also performed on a single biotite grain using a focused laser beam. Fifteen spots yield apparent ages ranging from 2016 ± 10 to 2152 ± 14 Ma, with a majority bracketed between 2116 and 2152 Ma (Fig. 3d). The age distribution does not correspond to a systematic core-rim zoning but it is noteworthy that the younger ages are from the mica rim, testifying of minor argon loss during cooling or later thermal disturbance. Again, the concordance of Pb/Pb and $^{40}Ar/^{39}Ar$ ages for this dioritic rock suggests that it experienced a rapid cooling history, making it a good candidate for frozen paleomagnetic directions at this time.

In the northern part of this area, the Degrad Roche orthogneiss (Degrad Roche, TAMP02 group) has been dated at 2131 ± 6 Ma (Table 1) on zircon with no apparent inherited isotopic component (Peucat in Delor et al., 2003a). Geological arguments (Hocquard and Schmitt, 1980), the relationships with surrounding formations, its well structured aspect and the isotopic data favour a syntectonic emplacement along a major tectonic corridor. Middle to fine grained granites, and sometimes porphyroidic ones, were sampled in the northern part (TAMP03 group) and in the southern part (TAMP05 group) of the Tampok River. They have in common to crosscut all other formations and to be quite isotropic. The northern outcrops sometimes contain some volcano-sedimentary relics while the southern ones are more often porphyroidic and contain some gabbro-dioritic relics. The southern granite was dated on zircon (ARA61, Table 1, TAMP05 group) at 2112 ± 2 Ma, the northern is younger than 2131 Ma and may have a similar or younger age. Two metamorphic rocks were sampled in the Tampok/Ouaqui area. In the vicinity of the Degrad Roche orthogneiss, some metavolcanics (TAMP01 group) are known to predate the major intrusions. However, their metamorphism may have occurred during the Degrad Roche orthogneiss syntectonic emplacement. In the southern part, gneissic rocks, dominantly paragneisses with sometimes amphibolites (TAMP07 group) are known to be intruded by all plutonic intrusions and therefore are older than 2155 Ma.

Of special interest for this study some argon results on the western Meso rhyacian domain (ARA 83, ITA 40) show amphibole and biotite ages similar to the U/Pb ages on zircons for the same rocks and therefore allow the use of paleomagnetic data for period greater than 2100 Ma, which has not been done in the previous studies (Nomade et al., 2002).

2.2.3. Northern, eastern and southern Neo Rhyacian main granitic magmatism (D2a sinistral shearing)

The Eastern Approuague (and Mataroni) River (Fig. 2c) is a major N–S collector which crosscuts the volcano-sedimentary formations to the North and gabbroic and granitic plutons to the South. The volcano-sedimentary metamorphic rocks consist of gneisses, amphibolites, sandstones and meta-mafic to felsic volcanic sequences. No age has been obtained to better constrain the emplacement of these formations and their metamorphism. It is likely that these events extended from 2200 to 2130 Ma as the various formations (Delor et al., 2003a) are crosscut by younger intrusions. MATA03, APPR01 and APPR04 groups belong to these metavolcano-sedimentary formations. For the plutonic intrusions, a set of Pb/Pb zircon results have

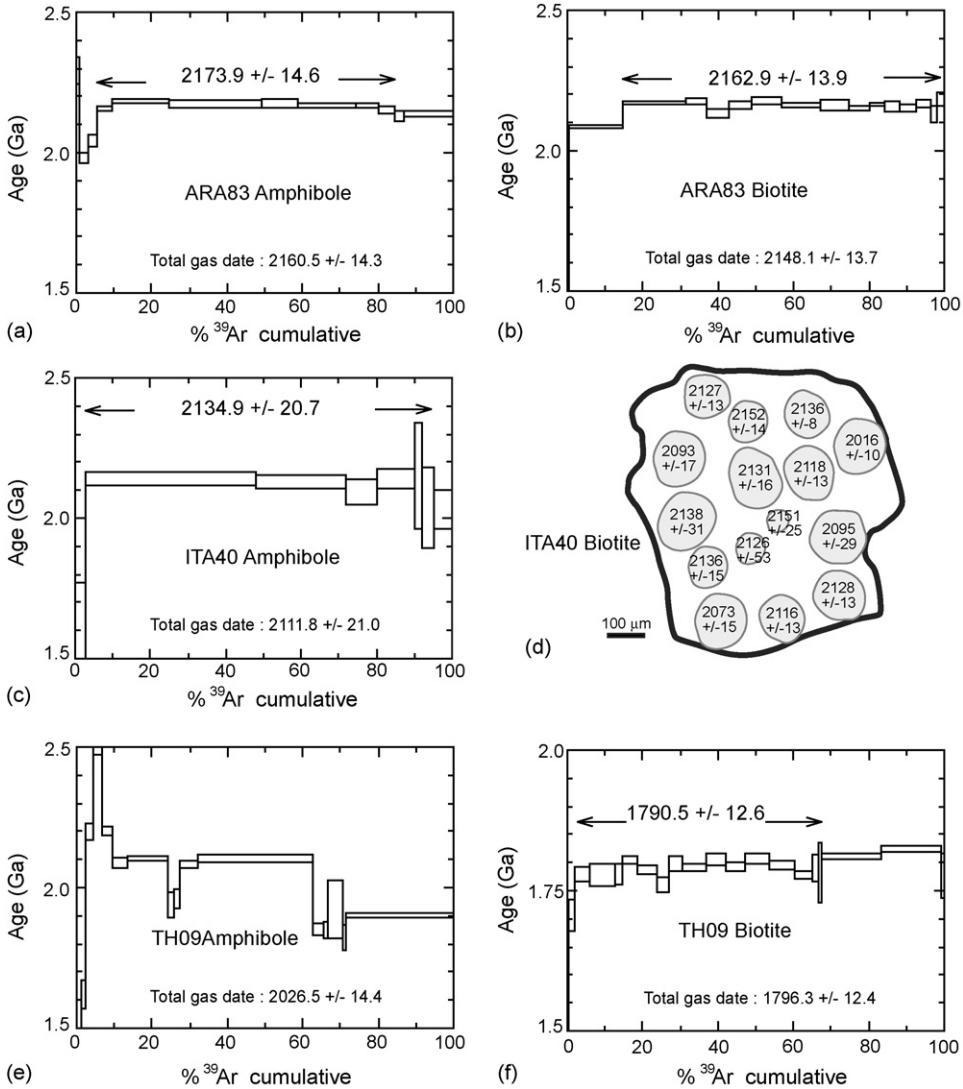


Fig. 3. (a–f) $^{40}Ar/^{39}Ar$ age spectra and spot fusion map for samples dated during this study.

been obtained (Vanderhaeghe et al., 1998; Delor et al., 2003a) which suggests the succession of a few pulses. The earliest pulse has been dated (REG27) at 2114.6 ± 2.5 Ma on a tonalite from the southern Mataroni River (MATA02 group; Vanderhaeghe et al., 1998). A slightly later pulse is well characterised by the intrusion of high-K granites around 2090 Ma with U/Pb zircon ages at 2093 ± 3.2 Ma (REG25; APPR02 group; Vanderhaeghe et al., 1998) and 2089 ± 4 Ma (DOC49; APPR06 group) along the Approuague River. The latest pulse is dated at 2084 ± 6 Ma on a leucogranite from the Mataroni area (REG26; MATA01 group; Vanderhaeghe et al., 1998). APPR05 and APPR08 groups are assigned to the 2093–2084 Ma time period.

A similar in age gabbroic intrusion emplaced along the Approuague River. This pluton has been dated (GUD36; APPR03 group) at 2094 ± 4 Ma and it intrudes the 2093 ± 3.2 Ma granite (APPR02 group). Field observations of the contact between both intrusions and the mafic and felsic nature of these rocks favour a similar crystallisation age at different temperatures, around 900 °C for the gabbro and 750 °C for the granite. If we consider they undergo a similar cooling history, they may have reached the temperature of acquisition of remanent magnetization, around 575–500 °C at different period. Similar late basic intrusion also exists in the Oyapock area (Nomade et al., 2001) with $^{40}Ar/^{39}Ar$ ages on amphibole (2020 ± 4 Ma) and on biotite (1973 ± 4 Ma) which were recently completed by a 2098 ± 2 Ma Pb/Pb

age on zircon (Delor et al., 2003a). This set of geochronological data suggests that this type of intrusion experienced slow cooling from 1000 to 300 °C over more than 100 Ma.

The Eastern Armontabo Mg–K granite (ARMO group, Fig. 1b) has been identified recently during a 1:100,000 scale detailed mapping program. It has been dated (ARM001) at 2080 ± 4.5 Ma (Lafon et al., 2003). This highly magnetite rich biotite granite is intermediate between the main 2100–2090 Ma granitic intrusions and the late 2070–2060 Ma metaluminous monzogranites. A biotite $^{40}Ar/^{39}Ar$ age of 1935.6 ± 12.1 Ma has been obtained on this intrusion (Enjolvy, 2004).

The Southern Tumuc Humac area (TUMU group, Fig. 1b) had never been geologically investigated before this geological mapping program (Delor et al., 2001) as no river neither road reach this area. We benefited from the heavy logistic of the French Army (2ème RIMA) to investigate around the "Borne 2" boundary stone which marks the frontier between French Guiana and Brazil. TUMU group corresponds to these Mg–K rich well structured granitic intrusions which have been dated at 2100 ± 1 Ma (TH09, Table 1). Biotite and amphibole from this sample have been dated (Table 2; Fig. 3e and f) using $^{40}Ar/^{39}Ar$ step-heating single grain laser probe experiments. Amphibole displays a discordant age pattern with apparent ages that decrease progressively during heating from 2488 to 1820 Ma, with a total gas date of 2027 ± 14 Ma (Fig. 3d). No plateau age can be derived from this sample. Biotite from the same sample has a less disturbed age spectrum with a plateau age of 1791 ± 13 Ma related to 65% of the argon released (Fig. 3e). The last portion of the spectrum displays somewhat older ages that increase up to 1825 Ma. Taken together, these $^{40}Ar/^{39}Ar$ ages indicate that this Mg–K rich granite has been potentially strongly overprinted by a thermal event around 1.8 Ga. It could be responsible for total argon resetting in biotite and only partial resetting in amphibole due to its higher retentivity for argon.

2.2.4. Northern Late Rhyacian metaluminous granitic magmatism (D2b dextral shearing)

The Northern metaluminous monzogranites (Fig. 1b) are only exposed on the northern edge of the sedimentary basins, along the coast, where they intrude the metasedimentary formations. They correspond (Delor et al., 2003a) to the latest identified plutonic pulse in French Guiana. An U/Pb zircon age of 2060 ± 4.1 Ma was obtained on a granite from the Petit Saut quarry (PESA group). A Pb/Pb age of 2069 ± 4 Ma was derived from the Organabo site (ORGA group). At the Roche Corail quarry (ROCO group), two U/Th/Pb ages on monazite at 2095 ± 6 and 2061 ± 15 Ma were obtained from a pegmatite which crosscuts the sampled monzogranite. The MATI group, although not dated, belongs to these late metaluminous intrusions from the Matiti quarry.

2.2.5. Neoproterozoic

In the southernmost sampled part of the Tampok River a doleritic dike crosscuts all formations. Supposedly Jurassic in age as most known doleritic dikes of French Guiana (Nomade et al., 2000), it was sampled for a baked contact test (Fig. 2b, BCT). The paleomagnetic results, discussed later, contrasted with expected Jurassic paleomagnetic directions. This led to date this late intrusion and a K/Ar age of 809 ± 29 Ma was obtained on whole rock (Delor et al., 2003a).

3. Paleomagnetic sampling and analyses

All encountered rock formations were sampled for rockmagnetic (anisotropy of magnetic susceptibility) and paleomagnetic purpose. We collected around 700 minicores from more than 110 sites using a portable petrol-powered drill. Other sampling has already been presented in Nomade et al. (2000, 2001, 2003). Standard 2.54 cm diameter cores were taken with lengths varying from 3 to 10 cm depending on rock hardness. Usually 3–10 cores were drilled per site, with duplication of sites along the rivers for similar lithologies. Cores were taken with varying azimuths to avoid systematic sampling errors and they were oriented with a magnetic compass and a solar compass whenever possible. A magnetic declination, according to the IGRF-95 or IGRF-2000 was used to correct all measured azimuths when solar orientation was missing. Magnetic and solar azimuths were usually similar at less than a few degrees.

All samples were sliced to standard size (2.2 cm long, 2.54 cm diameter) at BRGM. Left pieces of samples were crushed to powders for susceptibility/temperature experiments. All measurements were done at the BRGM-University of Orléans joint Laboratory (Laboratoire de Magnétisme des Roches). Magnetic remanence was measured with a spinner magnetometer (JR5, AGICO, Czech Republik). Demagnetization procedures were done using an automated 3-axis tumbler alternating field (AF) demagnetizer (LDA-3, AGICO, Czech Republik) or a Pyrox furnace. Isothermal remanent magnetization (IRM) were performed with an impulse magnetizer (IM10-30, A.S.C. Scientific, USA). Susceptibility versus temperature experiments were done with a CS3 appara-

tus coupled with a KLY-3 Kappabridge (AGICO, Czech Republik).

4. Rock magnetic and paleomagnetic results

4.1. Rock magnetism

All samples were subject to analysis of their Anisotropy of Magnetic Susceptibility (AMS) in order to compare the magnetic structural lineation and foliation with the geological ones as a possible tool for complementary structural observations for the geological mapping program. Most of the AMS directions showed complementary results as an efficient physical parameter for the structural geology mapping understanding. The anisotropy degree appeared to vary between low, less than 15%, and very high, up to 70%. Most of the plutonic samples belonged to the low group while the high group was dominated by the metamorphic rocks. Tentative correlation between the anisotropy degree, the anisotropy directions and the magnetic directions detailed hereafter appeared inconclusive. We therefore assume that the magnetic remanence has not been affected by the cooling structuration, when present, for the plutonic rocks. The metamorphic series, sometimes highly structured even though they show a low anisotropy degree, give rise to the same conclusion which could indicate that magnetic minerals were formed at the end of the metamorphism and therefore independently of the structuration.

In order to identify the main magnetic carriers of the studied samples, isothermal remanent magnetization (IRM) and low field susceptibility/temperature were performed on selected samples and powders from different representative lithologies. Examples of IRM acquisition curves (Fig. 4a and b) show similar behaviour with a rapid acquisition up to 0.1–0.2 T above which the saturation appears nearly complete although a slight increase in intensity is still noticed. This shows that magnetite is the dominant magnetic mineral present in the paleomagnetic samples while a low hematite content can be possible. These IRM interpretations are reinforced by the low field susceptibility/temperature experiments (Fig. 4c–h) which show the dominant presence of minerals from the titano-magnetite family marked by the sharp decrease around 550–580 °C. However, most samples still present variations up to 700 °C which could correspond to the presence of hematite. These experiments favour the dominance of minerals from the titano-magnetite family as the main magnetic carrier with a possible low contribution of hematite. This was already mentioned and detailed by Nomade et al. (2001, 2003).

4.2. Paleomagnetism

All samples were analyzed through thermal or alternating field techniques. Usually, thermal demagnetization appeared to be more efficient. However, some mafic rocks which had very high Natural Remanent Magnetization (NRM) intensities, probably due to lightening strikes, needed AF demagnetization to separate this secondary magnetization from the characteristic one. After treatment, all samples were represented on orthogonal (Zijderveld, 1967) vector plots for interpretation. Principal component analysis (Kirschvink, 1980) was used to characterize the magnetic components and mean directions were calculated using Fisher (1953) statistics.

On the following paragraphs and discussions the polarities will be arbitrarily chosen as normal when the inclination is positive and high and, when the inclination is low and the declination comprised around 270° and 360°. It will be taken as reversed when the inclination is negative and high and, when the inclination is low and the declination comprised around 90° and 180°. This will be used in order to minimize possible rotations and/or latitudinal displacements between groups (rock formations) of varying lithologies and/or ages on the discussion.

No clear mean secondary components were evidenced during this study, depending on the site location and its proximity with younger plutonic rocks and its degree of weathering. Possible secondary components were always largely scattered more or less around the Present Earth Field direction. This is therefore not discussed on the following paragraphs which concentrate on the identified and dominant high temperature components.

On the following paragraphs, ages assigned to paleomagnetic groups will be rock formation ages obtained on zircons by the Pb/Pb evaporation method or the U/Pb ion microprobe method as discussed on paragraph 1.2 (Table 1). We attributed the youngest age found on the dated rocks when zircon dating showed two ages corresponding to crystallization and inherited ages. Possible estimated magnetic ages based on the use of zircon crystallization data coupled with cooling ages derived from $^{40}Ar/^{39}Ar$ dating of amphibole and/or biotite will be only presented in Section 5. Magnetic ages relative to rock ages will therefore be discussed afterwards and separately.

4.2.1. Eorhyacian "Ile de Cayenne" metamorphic basement: CAYE01 and 02 groups

Samples were collected from the Bourda, Gosselins and Polissoirs beaches (Fig. 2a). The Bourda beach outcrop exposes gneisses intruded by a 9 m large Juras-

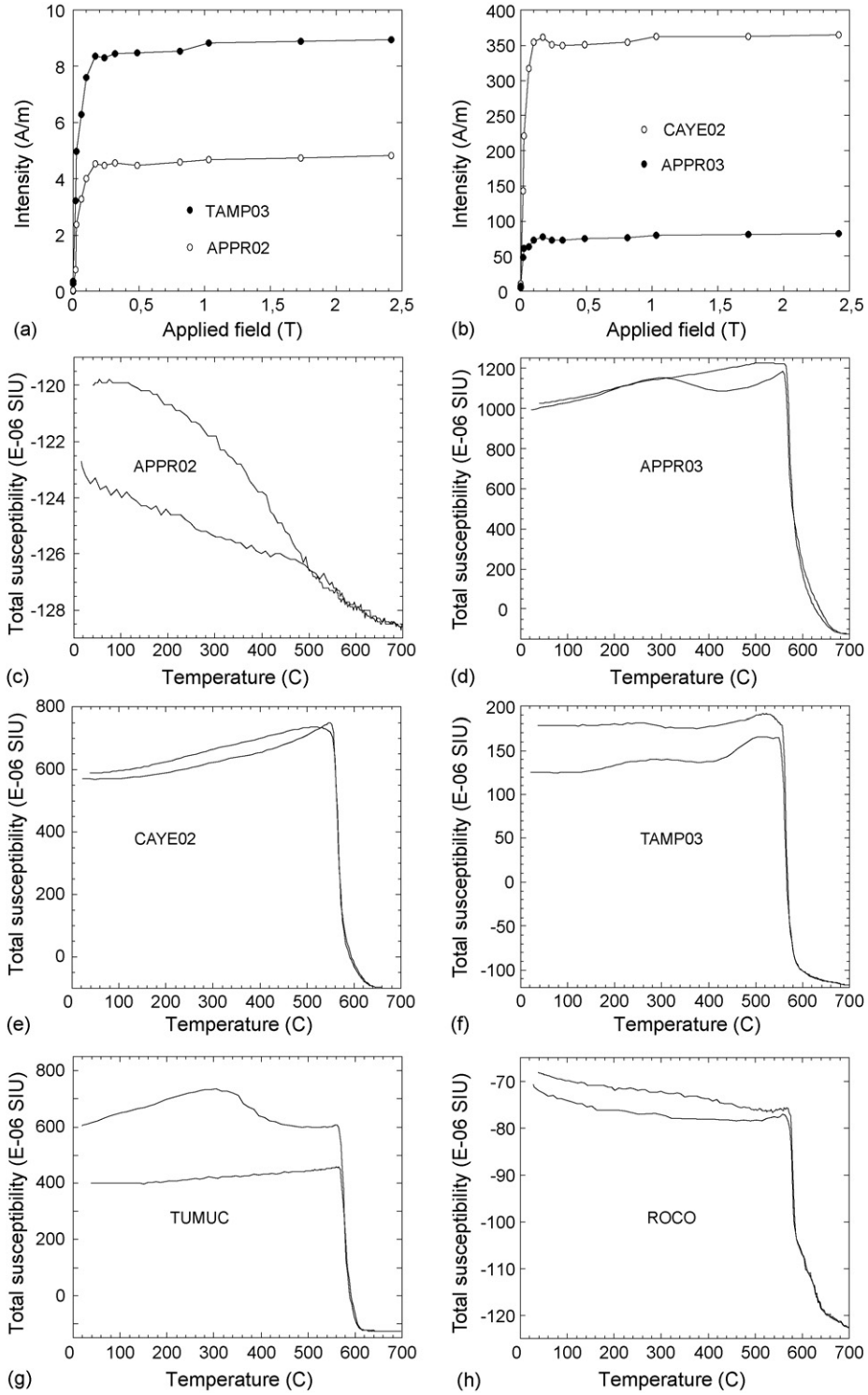


Fig. 4. Rock magnetic experiments. (a and b) Isothermal remanent magnetization curves of a few representative specimens; (c–h) magnetic susceptibility vs. temperature for some representative specimens.

sic dike. At the Gosselins beach, a 19 m large Jurassic doleritic dike crosscuts a granodioritic to tonalitic rock. A few samples were taken from a 1 m large and 5 m long wrapped up block within the dike and from the border to 6 m on each side of this NS oriented dike. At the Polissoirs beach, dioritic rocks are also present and they are intruded by gabbroic and doleritic dikes.

At the Bourda beach, samples behave similarly during thermal demagnetization but differ in direction relative to their distance from the dike. Samples being close to the dike (Fig. 5a) present a low unblocking temperature component of non significant direction above which a high temperature component can be defined between 550 and 585–595 °C. This component is only of normal polarity, northerly directed and with a shallow negative inclination similar to the dike one. For samples farther than 1 m from the dike (Fig. 5b), only one high temperature component is defined. It has a shallow inclination westerly or easterly directed. In Fig. 5c, directions from the gneiss are represented. Samples being below 1 m from the dike are clearly remagnetized samples due to the dike intrusion while samples away from 1 m define the CAYE01 group (Table 3). This baked contact test was already illustrated in Fig. 3b of Nomade et al. (2003). The age of the rocks which define this group direction has been established at 2173 ± 2 Ma on a tholeitic trondjhemite (CAY18A; Table 1; Vanderhaeghe et al., 1998).

At the Polissoirs beach, the diorite of the Mahury massif is intruded by gabbroic and doleritic dikes. After removal of a low unblocking temperature direction of non significant direction, a high temperature component can be defined between 550 and 585–595 °C. This component is different from the gabbro and dolerites ones as the inclination is shallow negative and the declination is slightly to the northwest (Fig. 5d). Apart from samples close to the dikes, all dioritic samples preserved a paleomagnetic signal clearly distinct and therefore older from these intrusive phases (Fig. 5e). This positive baked contact test between this Paleoproterozoic formation and the Early Jurassic dikes leads to the conclusion that only close to the dikes, samples are totally remagnetized but diorite samples taken away from the dikes preserved a paleomagnetic signal probably acquired during the Paleoproterozoic.

At the Gosselins beach, similar observations are made with sample directions. A high temperature component is clearly isolated from 500 to 580–600 °C on all samples of granodiorite. It has a shallow positive inclination and it slightly differs in declination between samples less than 2 m from the dike with a northeasterly declination and those located away from 2 m (Fig. 5f) with a north, slightly northwest declination. Representation of all the high temperature component directions from the Gosselins beach samples (Fig. 5g) clearly confirms the sample remagnetization up to 2 m from the dike. This baked contact test was already illustrated in Fig. 3a of Nomade et al. (2003).

Samples slightly away from the dikes at the Polissoirs beach and away from 2 m at the Gosselins beach are associated to define (Table 3) the Paleoproterozoic dioritic direction of the CAYE02 group. This group belongs to the gabbro/dioritic plutonic intrusion of the Mahury Mountain dated close to the Gosselins beach at 2144 ± 2 Ma (CAY12A, Vanderhaeghe et al., 1998; Table 1).

Paleomagnetic results obtained in the Cayenne area show strong evidence of a non pervasive influence of the Early Jurassic (Nomade et al., 2001) doleritic dike intrusions. Remagnetization only depends on the dike width. Directions found away from these dikes are probably representative of a Paleoproterozoic magnetization. In this area, the youngest age of 2144 ± 2 Ma refers to the CAYE02 group (diorite) while the 2173 ± 2 Ma age refers to CAYE01 group (gneiss).

4.2.2. D1 Western Mesorhyacian TTG-greenstone belt: TAMP01 to TAMP08 groups

The Western Tampok River is an EW river up to its continuation, the Ouaki River, where it turns to a NNW–SSE direction (Figs. 1b and 2b). It is one of the main collector flowing to the Maroni River which marks the boundary between French Guiana and Suriname. Fifty sites (280 samples) were sampled along a 15 km transverse on the Ouaki River and along 65 km on the Tampok River. As this area is smaller than the Approuague area (cf. Section 4.2.3) and as the lithologies vary frequently we combined distant sites of similar lithologies, on the basis of previous geological investigations (Hocquard and Schmitt, 1980). This allowed to avoid the creation of an excessive number of groups relative to those defined for other areas of French Guiana. Eight groups of combined sites/outcrops are presented, four groups are of metamorphic rocks, the others four of plutonic rocks. A site of a doleritic dike was also sampled to perform a baked contact test. This dike has been since dated, by K/Ar on whole rock, at 809 ± 29 Ma. Most of the outcropping plutonic rocks are dominantly of mafic affinity (gabbros, diorites and granodiorites). However, a few granitic exposures occurred for which it is clear they intrude the mafic formations. Detailed geological observations (Hocquard and Schmitt, 1980) and new isotopic data (Delor et al., 2003a) are in close agreement.

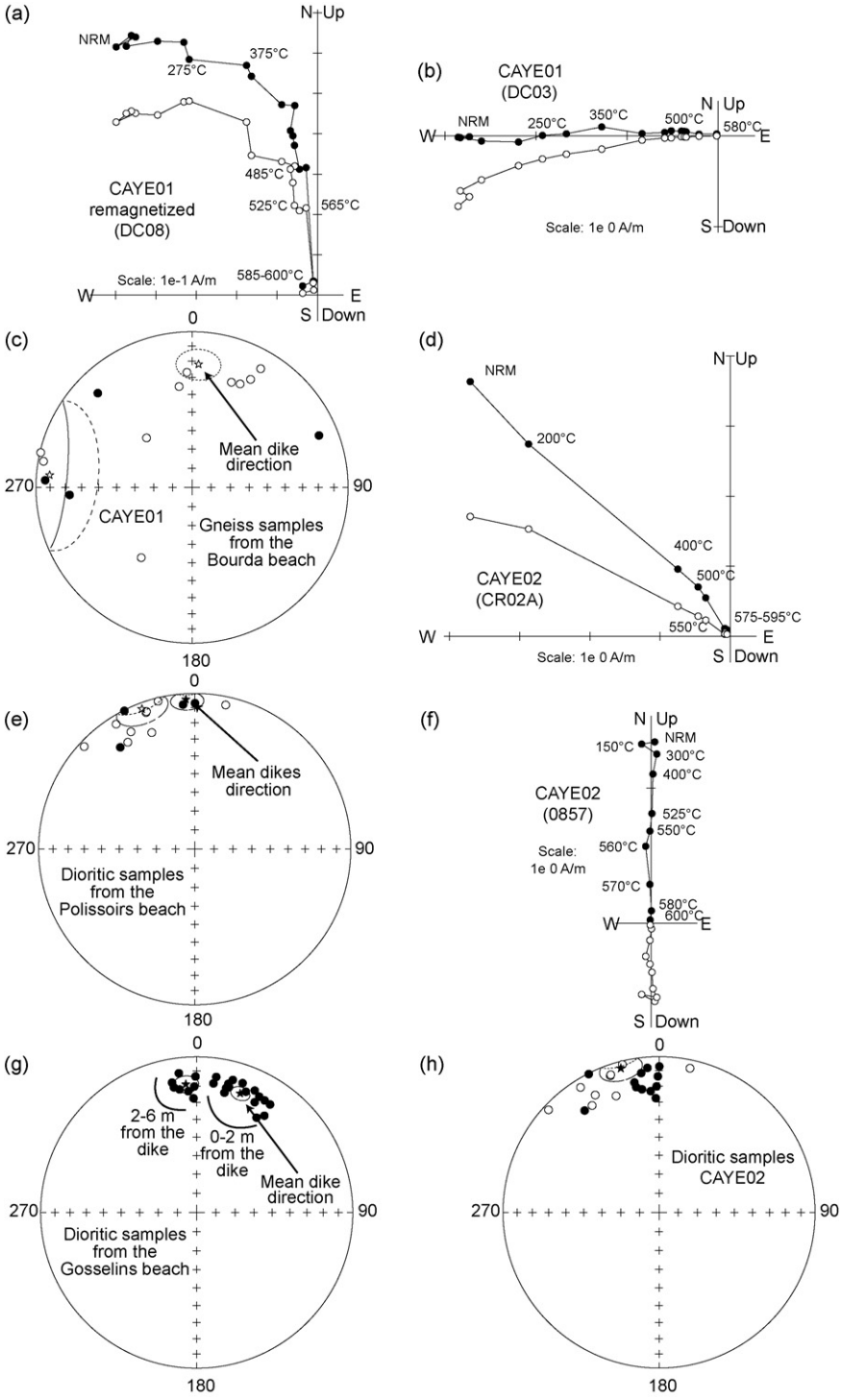


Fig. 5. Representative orthogonal vector diagrams for thermal demagnetization of specimens from the Cayenne Bourda beach area (a and b, CAYE01), full (empty) symbols represent horizontal (vertical) projections. Equal-area projection of directions from the Bourda beach (c) with indication of remagnetized samples, full (empty) symbols represent positive (negative) inclinations. (d and e) Same as (a and c) for samples from the Polissoirs beach and (f and g) for samples from the Gosselins beach. (h) CAYE02 directions isolated from dioritic samples of the Polissoirs and Gosselins beaches.

Table 3
Paleomagnetic mean directions for the different groups from the French Guiana Territory

Paleomagnetic group	Lithology	Latitude (°N)	Longitude (°E)	Sites Samples	*n*	Nor/Rev	Declination	Inclination	*k*	a95
Cayenne Island										
CAYE01	Gneiss	4.92	−52.28	1/14	8	7/1	275.0	−9.4	4.3	30.2
CAYE02	Diorite/tonalite	4.92	−52.28	2/39	21	21/0	345.2	4.8	15.3	8.4
Tampok and Ouaki rivers										
TAMP01	Metavolcanite	3.39	−53.86	6/31	14	12/2	310.7	12.3	7.8	15.3
TAMP02	"Ouaki" orthogneiss	3.40	−53.82	6/39	20	16/4	308.5	4.6	8.7	11.8
TAMP03	Granite	3.27	−53.81	8/47	36	26/10	210.9	84.0	9.4	8.2
TAMP04	Gabbro-diorite	3.28	−53.82	4/18	13	4/9	299.7	−24.6	15	11.1
TAMP05	Granite	3.16	−53.73	4/18	12	9/3	285.1	−7.5	5.9	19.5
TAMP06	Granodiorite	3.25	−53.77	5/31	25	14/11	357.6	−18.2	8.3	10.7
TAMP07	Gneiss-amphibolite	3.17	−53.70	10/53	31	16/15	283.8	−2.3	4.9	13
TAMP08	Orthogneiss	3.13	−53.68	4/15	10	8/2	352.1	−13.0	7.4	19.1
Mataroni and Approuague rivers										
MATA01	Granite	4.22	−52.22	3/13	9	6/3	287.0	26.9	3.4	33.1
MATA02	Tonalite	4.13	−52.20	1/6	7	5/2	301.8	79.0	8.2	22.5
MATA03	Metavolcanite	4.22	−52.18	4/20	6	1/5	248.6	56.6	10.8	21.4
APPR01	Metavolcanite	4.23	−52.25	2/8	9	3/6	306.9	15.8	12.4	15.2
APPR02	Granite	4.19	−52.33	6/40	35	22/13	272.4	85.6	7.2	9.7
APPR03	Gabbro	4.18	−52.34	4/26	32	27/5	137.3	67.0	18.2	6.1
APPR04	Metavolcanite	4.12	−52.40	3/23	7	3/4	273.9	−17.0	6.9	24.8
APPR05	Granite	4.04	−52.50	3/20	11	8/3	226.8	82.6	7.4	18.0
APPR06	Granodiorite	3.88	−52.58	4/20	15	10/5	209.7	76.4	10.5	12.4
APPR07	Granite	3.72	−52.64	3/10	8	7/1	300.5	−7.1	6.2	24.2
APPR08	Granite	3.67	−52.66	4/15	12	9/3	277.1	82.9	25.1	8.8
Tumuc Humac Mountains										
TUMU	Mg–K granite	2.18	−54.19	7/54	30	16/14	300.3	70.3	5.1	12.9
Armontabo river										
ARMO	Mg–K granite	3.68	−52.16	3/16	20	19/1	98.9	68.0	11.5	10.1
Coastal late monzogranite										
PESA	Granite	5.07	−53.03	1/10	7	6/1	327.5	−4.4	24.7	12.4
ROCO	Granite	5.10	−52.77	1/8	7	6/1	328.6	−2.7	15.5	15.8
MATI	Granite/Granodiorite	5.00	−52.58	2/11	8	5/3	329.3	−3.8	9.1	19.4
ORGA	Granite	5.55	−53.47	1/8	8	7/1	329.6	17.4	9.6	18.8

Latitude and longitude are the mean geographical coordinates for each group, sites/samples corresponds to the number of sites/outcrops and cores obtained, *n* is the number of specimens with the precision of their polarity (Nor/Rev); Declination; Inclination; *k*, precision parameter; a95, 95% confidence circle around the mean direction.

4.2.2.1. TAMP01 group. Six sites (31 cores) of metavolcanics mainly felsic rocks were sampled on the Tampok part lying between the Maroni and the Ouaki rivers (Fig. 2b). Thermal demagnetization applied on all samples reveals a clear high temperature component from 300–400 °C up to 660–685 °C (Fig. 6a). This component is probably carried by hematite although a small magnetite contribution is possible. Directions (Fig. 6b; Table 3) are dominantly of normal polarity with a shallow positive inclination and a NW declination. No age constraint exists for these rocks nor for their metamorphism. They are consequently coeval or older than the 2131 Ma old neighbouring orthogneiss (TAMP02 group) and possibly older than 2155 Ma which corresponds to the age of a granodioritic pluton (TAMP06 group) intruding similar volcano-sedimentary rocks southward.

4.2.2.2. TAMP02 group. Six sites (39 cores) were sampled along the Ouaki River on the "Ouaki Orthogneiss" (Fig. 2b). The NRM intensity does not vary significantly around the mean of 4.8 mA/m. A soft component is removed by 300–400 °C above which a high temperature component can be defined (Fig. 6c). This component is usually defined up to 575–585 °C but sometimes up to 635 °C implying a small hematite contribution to the magnetization. Directions are similar to those of TAMP01 group but a few reverse polarity directions are observed (Fig. 6d). The age of this

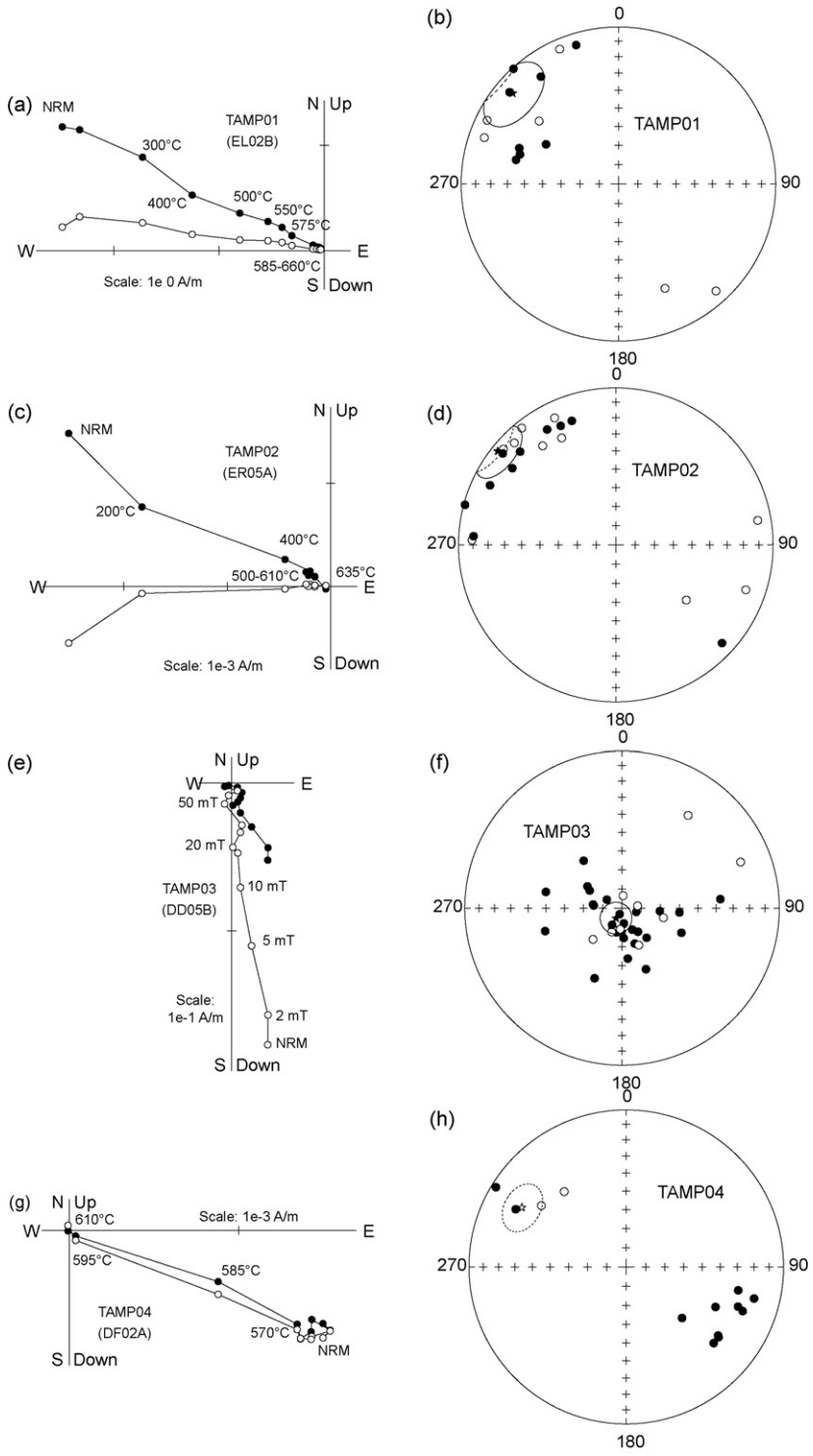


Fig. 6. Representative orthogonal vector diagrams for thermal or AF demagnetization of specimens from the different groups of the Tampok and Ouaki rivers and equal-area projections of the characteristic group directions: (a and b) TAMP01, (c and d) TAMP02, (e and f) TAMP03, (g and h) TAMP04, (i and j) TAMP05, (k and l) TAMP06, (m and n) TAMP07, (o and p) TAMP08 and baked contact test BCT (q–u). Symbols are the same as in Fig. 5.

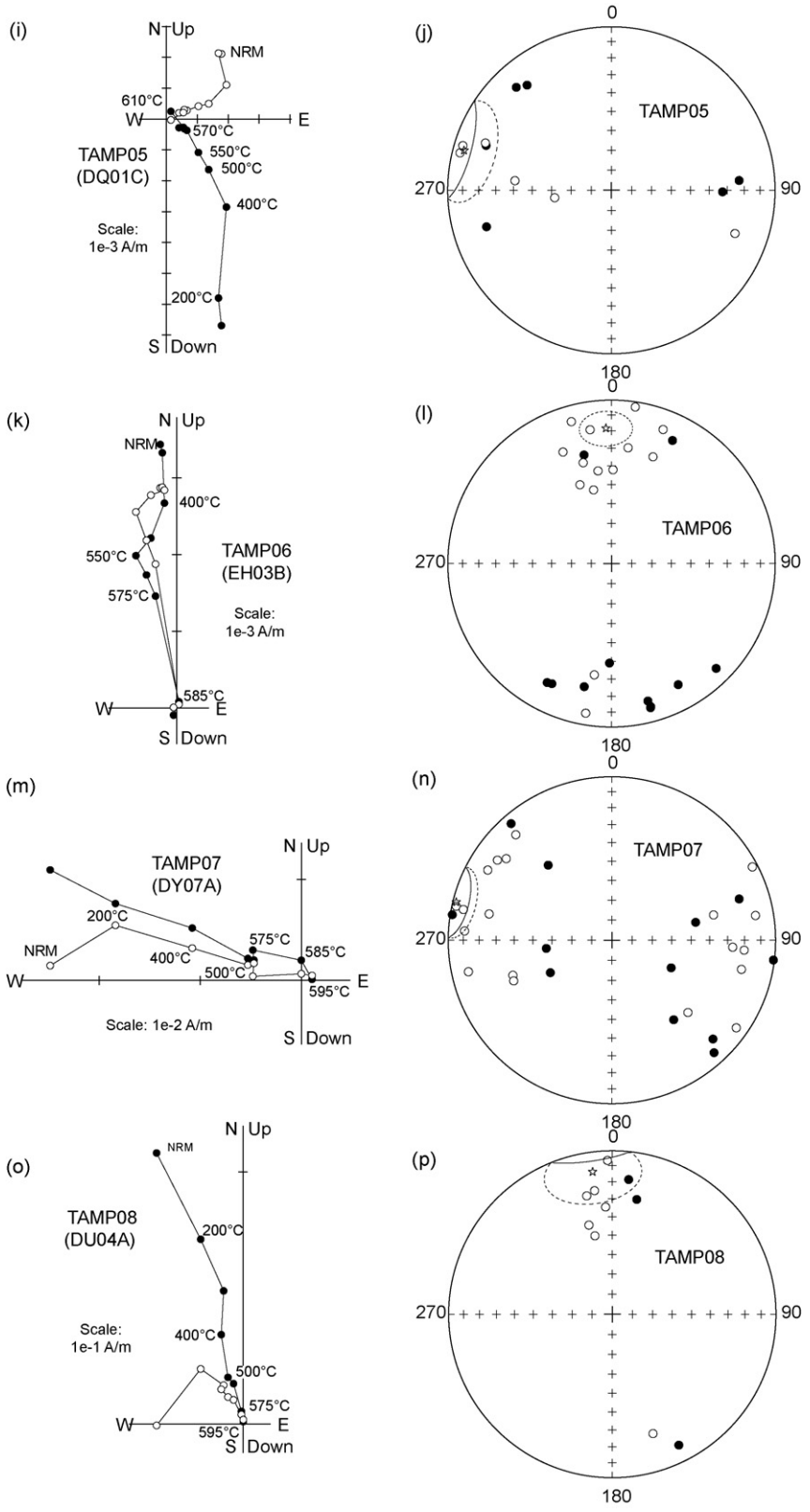


Fig. 6. (*Continued*)

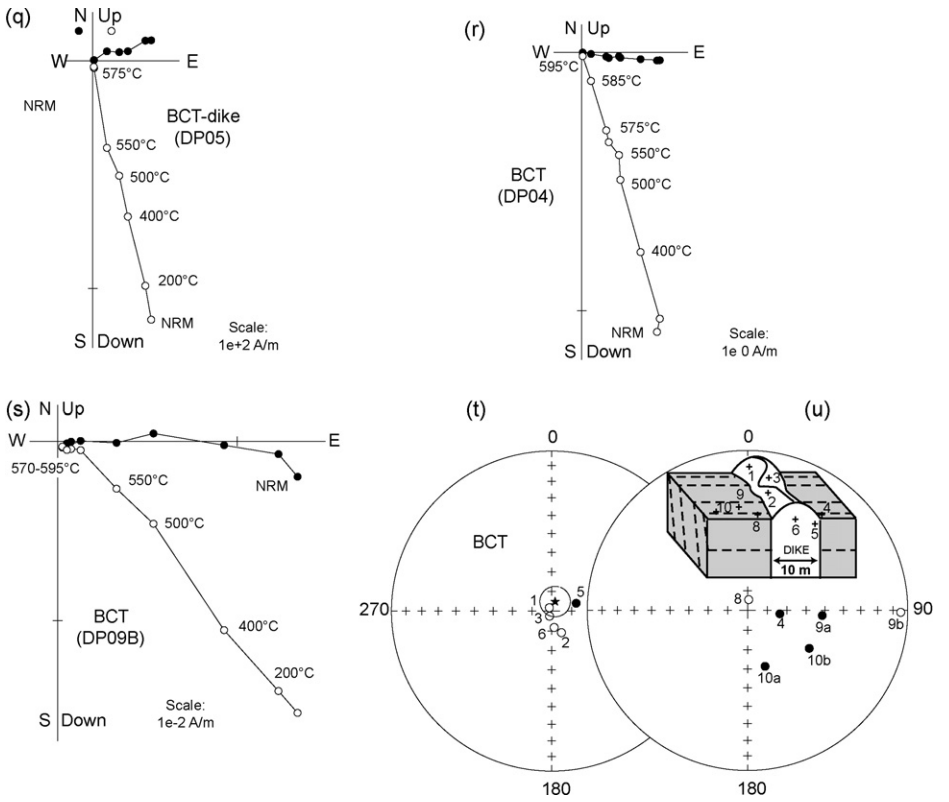


Fig. 6. (*Continued*).

rock, at the Degrad Roche site, is 2131 ± 6 Ma (Peucat in Delor et al., 2003a). The contact between the orthogneiss (TAMP02 group) and the metavolcanics (TAMP01 group) is not clear but the metamorphism of these metavolcanics is supposed to have occurred nearly at the time of the orthogneiss emplacement (Hocquard and Schmitt, 1980).

4.2.2.3. TAMP03 group. Eight sites (47 cores) were taken from small granitic occurrences and microdioritic to microgranitic veins. These are mainly concentrated on the northern part of the Tampok area (Fig. 2b) where small granitic plutons intrude the other formations (Hocquard and Schmitt, 1980; Marot, 1988). As the lithologies present a few variations, the NRM intensity varies between 0.1 mA/m and 3.3 A/m. Thermal or AF demagnetization (Fig. 6e) reveals the presence of a high temperature or high coercivity component of steep inclination. Both polarities are present and a mean direction was calculated with a high inclination (Table 3; Fig. 6f). These granites are middle to fined grained, dominantly isotropic. They correspond to a late phase younger than the emplacement of the well structured Ouaki orthogneiss (TAMP02 group). Paleomagnetic directions are consistent with the results obtained on 2093–2084 Ma old rocks from the Approuague River (cf. Section 4.2.3).

4.2.2.4. TAMP04 group. This group is from dioritic and gabbroic rocks (4 sites, 18 samples) of the major gabbro-dioritic intrusion identified in the Tampok area (Fig. 2b). These were dated at 2147 ± 5 Ma 15 km to the west of the same dioritic rocks (ITA40A, Table 1). Demagnetization reveals two different behaviours with a first constant and linear decay from 200 °C up to 575 °C for some samples and with no demagnetization up to 550–570 °C followed by a rapid decay up to 595–610 °C (Fig. 6g) for the others. Directions are mostly of reverse polarity with a shallow positive inclination and a southeast declination (Fig. 6h; Table 3). The similarity of Pb/Pb zircon, Ar/Ar amphibole and Ar/Ar biotite ages (Table 2) suggests that this TAMP04 group rocks experienced fast cooling and that their magnetic age nearly coincides with their emplacement age.

4.2.2.5. TAMP05 group. Four sites (16 cores) of granitic occurrences were sampled (Fig. 2b). They are recognized to be one of the latest observable plutonic

pulse as they intrude the other formation (Hocquard and Schmitt, 1980; Marot, 1988). NRM intensities are quite low usually around 2 mA/m. After removal of a low unblocking temperature component of varying although close to the present day field direction, a high temperature component is evidenced above 400 °C and up to 600 °C (Fig. 6i). Directions are of low inclination with an east or west declination with both polarities but moderately scattered (Fig. 6j; Table 3). These granitic occurrences were supposed to be similar to those of TAMP03 group. As the directions differed, one sample was dated which revealed an older Pb/Pb zircon age of 2112 ± 2 Ma (ARA61, Table 1) which confirms they are in time and therefore paleomagnetically different from the 2093–2084 Ma TAMP03 group.

4.2.2.6. TAMP06 group. Five sites (31 cores) of mainly granodioritic rocks were sampled (Fig. 2b). The mean NRM intensities is of 89.8 mA/m. After removal of a low unblocking temperature up to 200 or 400 °C, a high temperature component is evidenced up to 570–585 °C (Fig. 6k). Directions from all samples are moderately scattered around a mean direction of shallow negative inclination and northerly directed declination (Fig. 6l; Table 3). Both polarities are also encountered. This granodioritic body (ARA83, Table 1) is intruded by the gabbro-dioritic massif of the TAMP04 group dated at 2147 ± 5 Ma. A U/Pb zircon age of 2155 ± 3 Ma obtained at one site confirms the field relationships and therefore it is assigned to this group. Moreover, this intrusion is likely to have undergone a rapid cooling and no later pervasive perturbation as evidenced by $^{40}Ar/^{39}Ar$ age on amphibole at 2174 ± 15 Ma and on biotite at 2163 ± 14 Ma (Fig. 3a and b; Table 2).

4.2.2.7. TAMP07 group. Ten sites (53 cores) of gneisses and amphibolites were used to define this group (Fig. 2b). The weakest intensities are found on the amphibolite samples (mean 1.75 mA/m), gneiss samples having medium intensities (mean 87.7 mA/m). Samples behave differently from one site to the other but all show a clear high temperature component from 500 to 585–595 °C (Fig. 6m). Directions are highly scattered around the mean of shallow inclination and west slightly north-west declination (Fig. 6n; Table 3). The age of this group is older than 2155 Ma as the TAMP06 group granodiorite crosscuts this formation.

4.2.2.8. TAMP08 group. Four sites (15 cores) were taken from orthogneissic rocks (Fig. 2b). The NRM intensities are higher than those of similar lithology with a mean at 142 mA/m. Only one component is defined between 200 and 575 °C (Fig. 6o). This component is of both polarities (Fig. 6p) and very similar to the one of TAMP06 group (Fig. 6l; Table 3). This southern orthogneiss seems coeval to the granodiorite of TAMP06 group on geological grounds (Hocquard and Schmitt, 1980). Their paleomagnetic directions are also similar and an age of 2155 ± 3 Ma can be confidently assigned to this group.

4.2.2.9. Neoproterozoic Tampok dolerite. One site (5 cores) of a long (150–200 km) N140 doleritic dike (Fig. 2b) was sampled with its continuity into metasedimentary rocks (5 cores). This dolerite represents the youngest geological event found on the Tampok River, now dated at 809 ± 26 Ma (Delor et al., 2003a), which crosscuts all observed Paleoproterozoic rocks (Marot, 1988). This was the only exposure for a possible baked contact test along the Tampok River. The metasedimentary rocks belong to those already discussed for group TAMP07. On the dike samples, only one component, of steep inclination, is identified (Fig. 6q). Samples from the metasedimentary rocks show different behaviours. Samples close to the dike (Fig. 6r) show only one component, similar to the dike and characteristic of a complete remagnetization. However, samples away from a few meters (Fig. 6s) present two components, with a similar to the dike component followed by a high temperature component of less steep to shallow inclination, similar to the one identified for TAMP07 group. One sample of the dike (BCT) from the fine grained eastern border (Fig. 6t and geological inset) is of normal polarity while the others (coarse grained) are of reverse polarity. This shows that the cooling time was sufficiently long to record a reversal, from the fine grained border to the coarse grained centre and therefore cancel any possible secular variation effect. For the samples of the metasedimentary bedrock, we can observe that one sample (Fig. 6u, no. 4, geological inset) close to the normally magnetized border presents the same direction. The other samples present directions similar (Fig. 6u, no. 8) or different (Fig. 6u, nos. 9 and 10) when we move away westward from the dike. These directions tend toward those of TAMP07 group from which they lithologically belong. From these observations, we can conclude that no pervasive remagnetization occurred during the dike emplacement except on its close vicinity and probably all Proterozoic rock formations from this area have not been remagnetized during this dike emplacement. Detailed results from the dike are not presented on Table 3 as it does not belong to the Paleoproterozoic formations presented in this paper.

4.2.3. D1 to D2a Eastern Meso to Neo Rhyacian plutonic and metamorphic rocks: MATA01 to 03, and APPR01 to 08 groups

Along the Eastern Approuague and Mataroni rivers, 42 sites were sampled, 5 corresponding to Early Jurassic doleritic dikes (Nomade et al., 2000). Nine sites were from metamorphic rocks which defined three groups, 28 sites were from plutonic rocks which defined eight groups on the basis of similar lithology, ages and/or geographical proximity. Three groups were defined on the Mataroni River along a 15 km transverse while eight groups were on the Approuague River along a 110 km N–S transverse (Fig. 2c).

4.2.3.1. MATA01 group. Three sites (13 cores) of granites were sampled (Fig. 2c). Demagnetization is more or less noisy (Fig. 7a) probably due to the low magnetic content of these leucogranites. Selected directions are highly scattered (Table 3; Fig. 7b) although the inclinations are quite low. These rocks correspond to a leucogranite dated at 2084 ± 6 Ma (Table 1, REG 26, Vanderhaeghe et al., 1998).

4.2.3.2. MATA02 group. Only one site was taken from this dated tonalitic intrusion (Table 1, REG 27, Vanderhaeghe et al., 1998) which covers the southeastern part of the Mataroni River (Fig. 2c). Mean NRM intensities are around 0.5 A/m and the demagnetization is straightforward up to 575–590 °C (Fig. 7c) with no apparent secondary magnetization. As this outcrop is far from the closest younger granite outcrop (MATA01 group), later intrusions may not have affected this granite which probably kept a characteristic magnetization. Both polarities are present. The mean direction (Table 3; Fig. 7d) is of high inclination. This group has an age of 2115 ± 3 Ma although these rocks have inherited zircons at 2129 ± 3 Ma (Vanderhaeghe et al., 1998).

4.2.3.3. MATA03 group. This group is defined on metavolcanic rocks (Fig. 2c) close to the granitic intrusions or to some Early Jurassic dikes. Most samples were taken in contact with these intrusions. No high temperature component could be isolated from samples close to granites. On samples far away from intrusions (Fig. 7e), the high temperature component is usually defined between 400 and 575 °C. Isolated directions (Fig. 7f; Table 3) distinct from the dike, the MATA01 or the MATA02 directions have a medium inclination and a nearly east or west declination (Fig. 7f; Table 3). They are supposed to be representative of the magnetization acquired during or at the end of the metamorphic event which preceded the different intrusions. This group is therefore older than around 2115 Ma and no better constrained than being between 2200 and 2130 Ma according to Delor et al. (2003a).

4.2.3.4. APPR01 group. Two sites (8 cores) were taken from metavolcanic rocks (Fig. 2c). The initial NRM intensities do not vary significantly around the mean of 1.5 mA/m. Only one component appears during thermal demagnetization with a linearly decay, although slightly noisy, between the NRM and 575–590 °C (Fig. 7g). Components of both polarities are found (Fig. 7h) defining a west/northwest direction of shallow inclination (Fig. 7h; Table 3). This group is also no better constrained than being between 2200 and 2130 Ma according to Delor et al. (2003a).

4.2.3.5. APPR02 group. Six sites (40 cores) were taken from dated granitic rocks (Fig. 2c; Table 1). After removal of a low unblocking temperature component up to 200–300 °C (Fig. 7i) most samples linearly decay up to 575–595 °C defining one direction of high inclination (Table 3; Fig. 7j). Both polarities are found on the different outcrops and the mean direction is well clustered around 85° of inclination. The age of 2093.0 ± 3.2 Ma assigned to this group comes from one of our sites of this granite (Table 1, REG25, Vanderhaeghe et al., 1998).

4.2.3.6. APPR03 group. Four sites (26 cores) of mainly gabbroic rocks were sampled slightly south of the APPR02 group (Fig. 2c). The NRM intensities are high varying between 2.5 and 5.5 A/m. Some samples show only one direction after demagnetization while others display a possible low temperature secondary magnetization close to the Present Earth Field direction (Fig. 7k). The high temperature component has been defined between 525–550 and 590–615 °C which suggests a possible small hematite contribution upon the magnetization. Directions (Table 3; Fig. 7l) are well defined and dominantly of normal polarity. They are similar, although slightly different from the granite ones of APPR02 group which is intruded by this gabbro. However, this could not be used as a baked contact test as their ages and directions are close to each other (Table 3). An age of 2094 ± 4 Ma (GUD 32, Table 1) was obtained on a sample collected at a paleomagnetic site (Table 1) of this gabbro-dioritic intrusion and is therefore assigned to this group.

4.2.3.7. APPR04 group. Three sites (23 cores) were taken from metavolcanic rocks (Fig. 2c). Initial magnetic parameters present great variations for the inten-

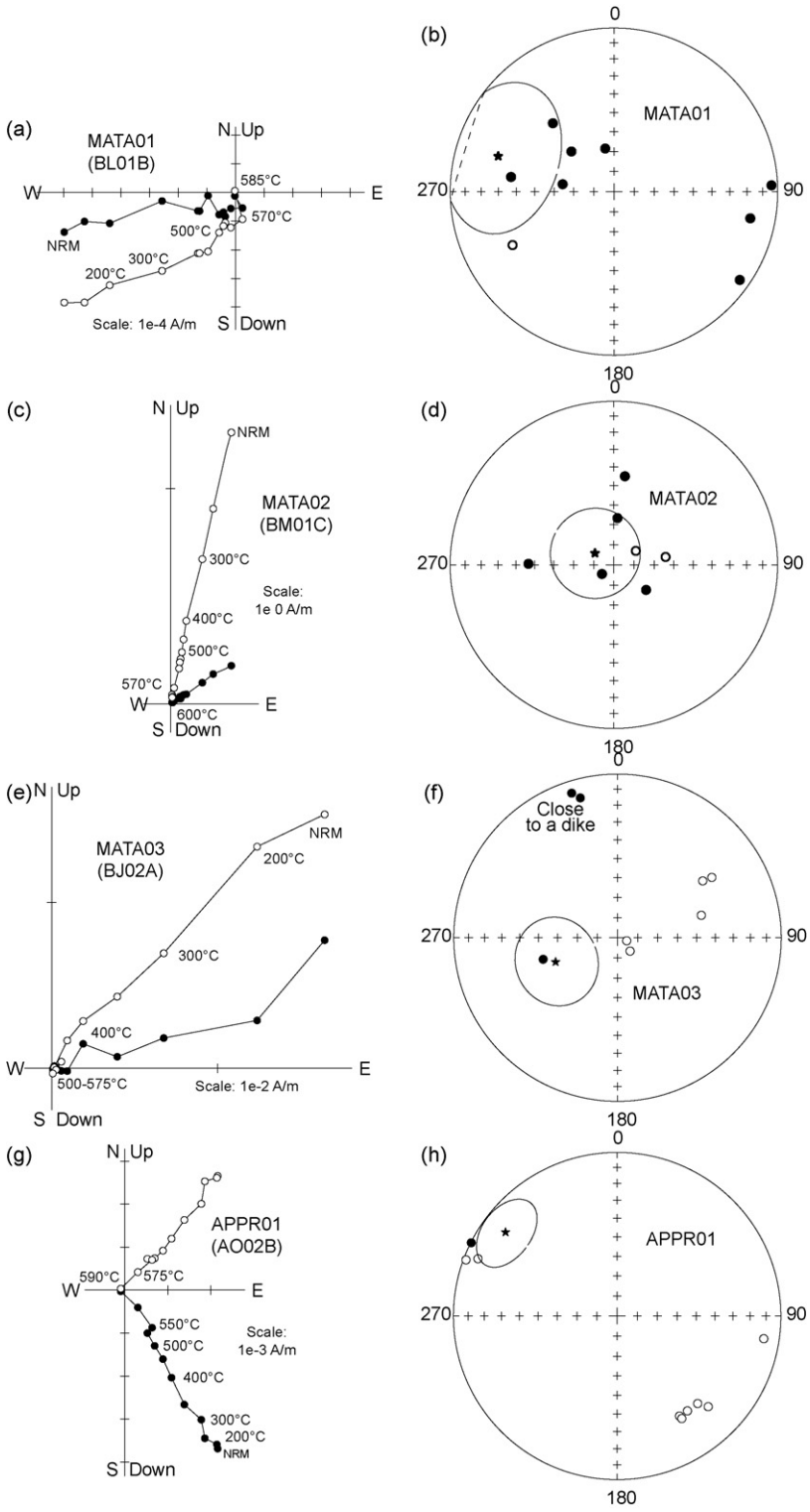


Fig. 7. Representative orthogonal vector diagrams for thermal or AF demagnetization of specimens from the different groups of the Approuague and Mataroni river and equal-area projections of the characteristic group directions: (a and b) MATA01, (c and d) MATA02, (e and f) MATA03, (g and h) APPR01, (i and j) APPR02, (k and l) APPR03, (m and n) APPR04, (o and p) APPR05, (q and r) APPR06, (s and t) APPR07, (u and v) APPR08. Symbols are the same as in Fig. 5.

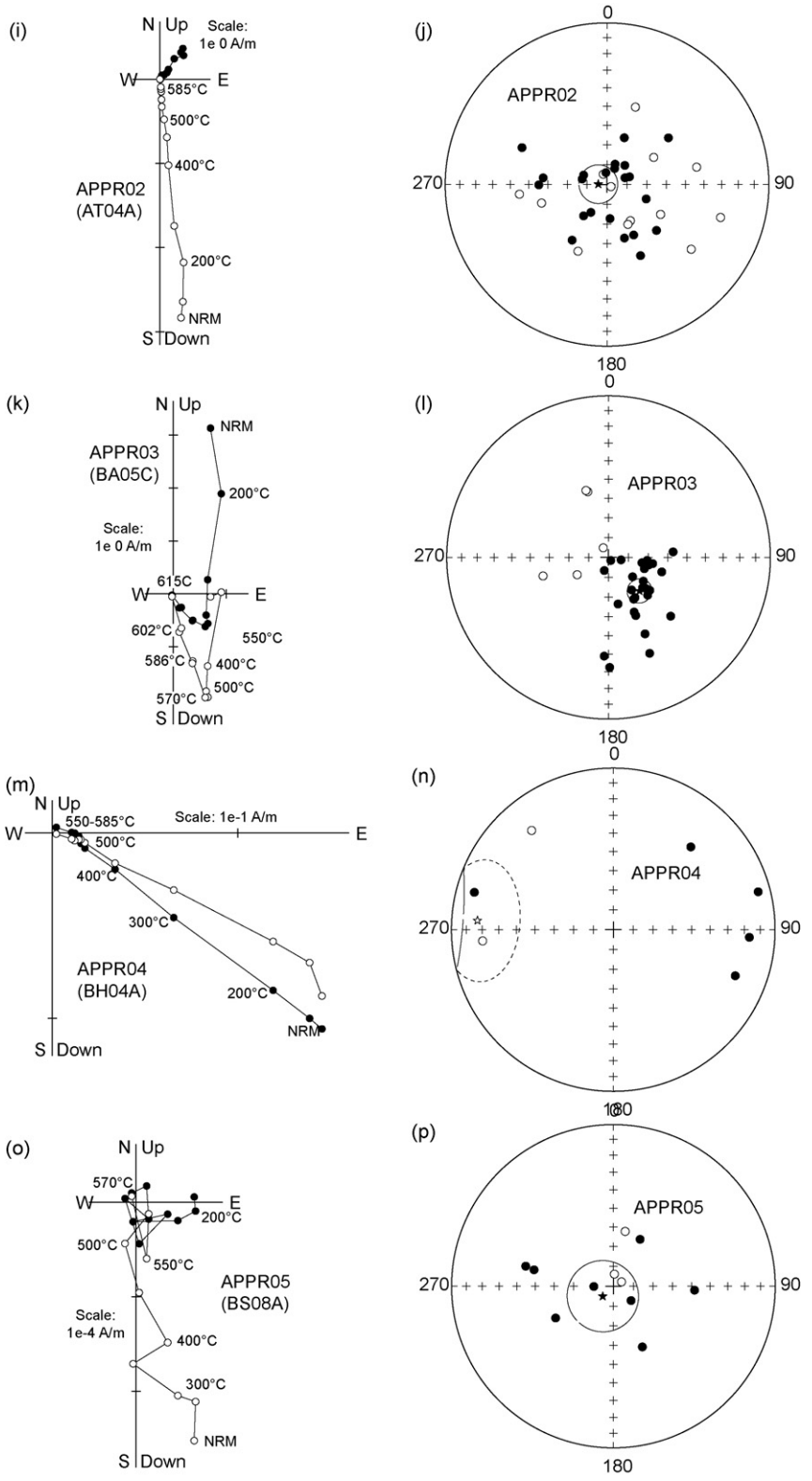


Fig. 7. (*Continued*)

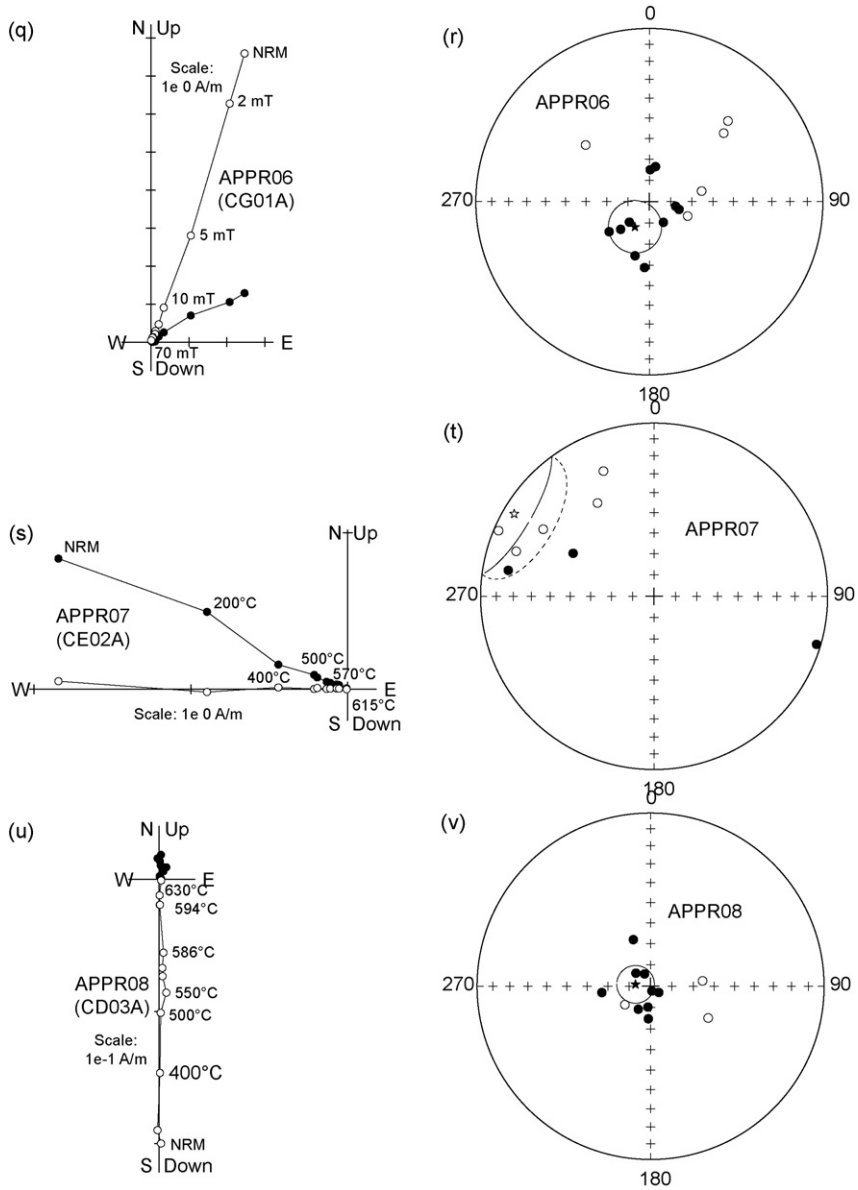


Fig. 7. (*Continued*).

sity (0.19 mA/m to 31.6 A/m). This intensity drops to low values above 300–400 °C. No representative direction is associated. From 500–525 to 550–585 °C, a high temperature component (Fig. 7m) appears. It has a shallow inclination and a west or east declination. As this direction (Fig. 7n; Table 3) is distinct from the surrounding granite directions, it can be representative of the magnetization acquired before the intrusion. This is consistent with an age estimate no better constrained than between 2200 and 2130 Ma (Delor et al., 2003a,b) like those (MATA03 and APPR01 groups) of similar lithology.

4.2.3.8. APPR05 group. Three sites (20 cores) of granodiorite were sampled along the Approuague River (Fig. 2c). NRM intensities are quite low compared to the other groups and demagnetization appears to be noisy (Fig. 7o). However, characteristic directions can be isolated from this group giving a mean direction similar to the one encountered on other groups along this river (Table 3; Fig. 7p). An age around 2093–2089 Ma may therefore be assigned to this group as it does not differ paleomagnetically from the other dated and similar plutonic groups (APPR02 and APPR06 groups).

4.2.3.9. APPR06 group. This group corresponds to granitic occurrences around 10 km south of the APPR05 group (Fig. 2c). Four sites (20 cores) were sampled. NRM intensities present strong variations between 60 mA/m and 8 A/m. There is usually only one direction (Fig. 7q) above 200 °C or 10 mT. This direction is of both polarities and of high inclination (Fig. 7r; Table 3). The direction is similar to the one encountered on granites, gabbros and granodiorites of the previous groups with an age of 2089 ± 4 Ma (DOC49, Table 1) obtained from a sample collected at one paleomagnetic site of this group.

4.2.3.10. APPR07 group. This group has been defined on paleomagnetic grounds. The lithology is similar to the APPR06 and 08 groups located north and south of these outcrops (Fig. 2c). However, the paleomagnetic results (Fig. 7s and t; Table 3) appear to be quite different with only a low inclination direction towards the NW. Moreover they also differ by their high intensity around 1–2 A/m which suggests they probably correspond to another granitic pulse of different composition and unknown age. Unfortunately only paleomagnetic samples were taken from the sites used to define this group and no age could be investigated. However, the sampled rocks, although similar to those of group APPR06 and 08 have the peculiarity to be granites rich in amphibolite enclaves. This is a characteristic of older intrusions found on other areas for which an age around 2150–2140 Ma is assigned on the new French Guiana geological map (Delor et al., 2001, 2003a).

4.2.3.11. APPR08 group. Four sites (15 cores) of granites were sampled on the southernmost accessible part of the Approuague River (Fig. 2c). The NRM intensities vary between very low (0.9 mA/m) to medium (0.3 A/m). Only one direction is evidenced during the demagnetization (Fig. 7u) and both polarities are found (Fig. 7v). These results are similar in lithology, intensity and paleomagnetic direction (Table 3) to those of APPR06 group located 20 km to the north. An age around 2089 Ma may therefore be attributed confidently to this group.

4.2.4. D2a Neo-Rhyacian Armontabo River granite: ARMO group

Three sites (16 cores) of highly magnetic (field susceptibility on outcrops around 50 SIU) Mg–K granite were sampled along the Eastern Armontabo River (Fig. 1b). At one site, a doleritic dike crosscuts the granite and allows performing a baked contact test. The NRM intensities are high, similar to those of the APPR03 gabbro intrusion with a mean at 2.98 A/m. Most of these samples were AF demagnetized after a first thermal step (Fig. 8a–d) done in order to highlight a possible goethite contribution. On the dike samples, only one component is identified (Fig. 8a). It has a shallow inclination and a N–NE declination. No sample in contact with the dike could be sampled but samples 2 m away (Fig. 8b) showed two components with one low coercivity direction similar to the dike one. A second, high coercivity component (Fig. 8c), of higher inclination and nearly east declination followed. This direction was similar to the one identified on other samples from granitic occurrences far from the dike (Fig. 8d). Directions (Fig. 8e) are dominantly of high positive inclination with an eastward declination. They differ significantly (Fig. 8e) from the doleritic dike directions which are similar to those encountered on Early Jurassic dikes (Nomade et al., 2000). This granite has been dated (Table 1) at one site with an age of 2080 ± 4 Ma on zircon. A 150 My younger age on biotite was derived by the $^{40}Ar/^{39}Ar$ method (Enjolvy, 2004).

4.2.5. D2a Southern Neo-Rhyacian Tumuc Humac Mg–K granite: TUMU group

Seven sites (54 cores) were sampled around the Southern "Borne 2" boundary stone between French Guiana and Brazil (Fig. 1b). The sampled Mg–K granites outcropped only on small streams. When rocks were present, particular care was taken in sampling the best exposed occurrences in order to avoid any possible local tilting (boulders) which could prevent any paleomagnetic analysis. The magnetic intensities are the highest observed on granitic rocks from French Guiana, similar to those of the Armontabo granite, with variations between 0.3 and 12 A/m. After removal of a low unblocking random secondary component, most samples show (Fig. 8f and g) a high temperature component, up to 595 °C, of high inclination and both polarities. Selected directions (Fig. 8h) enable us to obtain a mean direction (Table 3) for the Tumuc Humac mountain. From the 54 cores, only 30 specimens were used. One site (7 cores) of highly structured granite with up to 73% of anisotropy degree which produced outlier directions was rejected. For the other samples, we rejected those belonging to blocks which produced directions away for more than 30° from the mean and were therefore probably tilted. Other samples with no field evidence of tilting were kept, even if they had directions far from the mean. The mean direction is of high inclination and N–NW or S–SE declination. The age of 2100 ± 1 Ma obtained at one of our sites (TH09, Table 1) is assigned to this group. At this stage, the paleomagnetic results do not record the influence of a potential thermal event suggested by the young $^{40}Ar/^{39}Ar$ ages obtained on amphibole and biotite (Fig. 3e and f; Table 2).

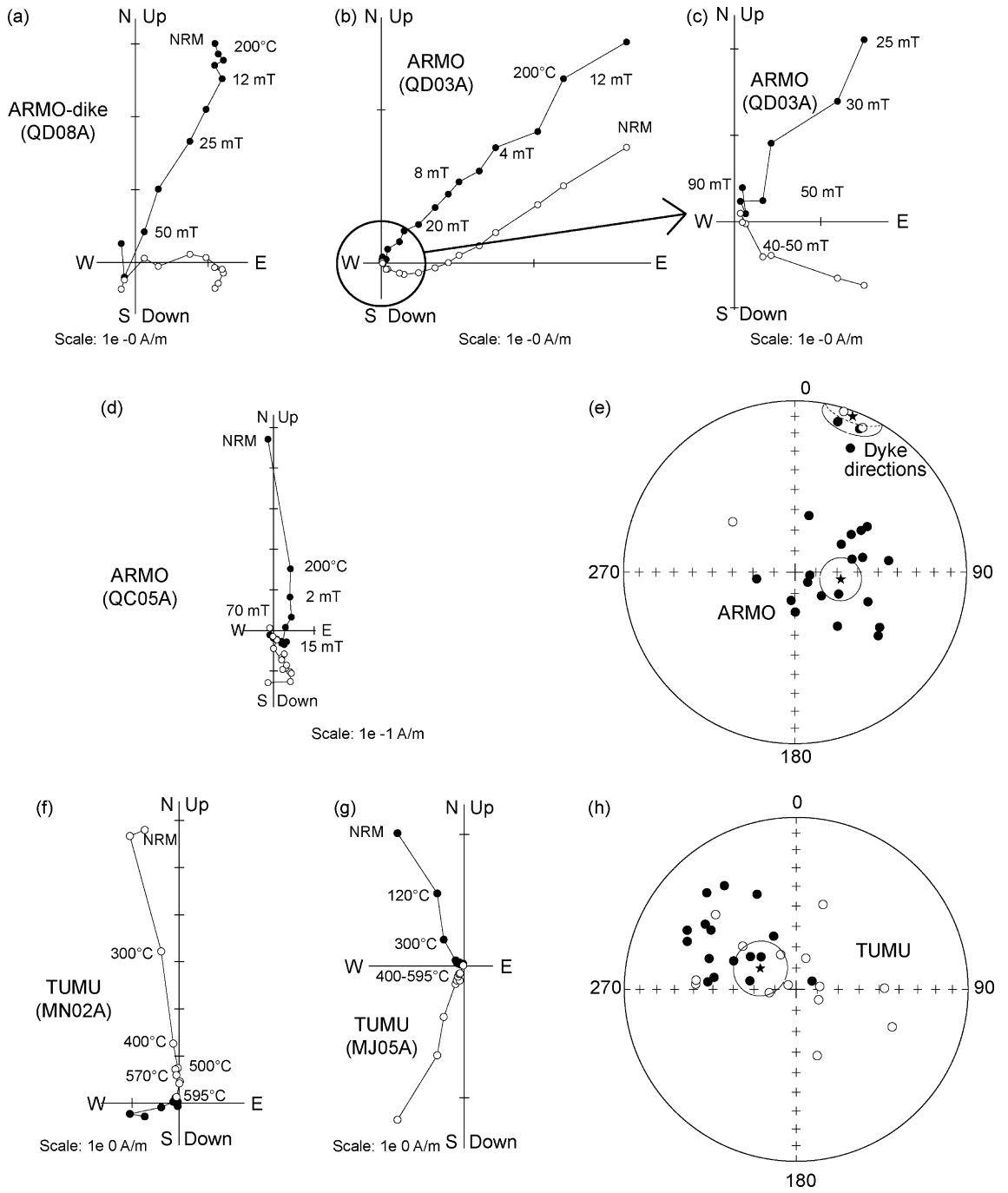


Fig. 8. Representative orthogonal vector diagrams and equal-area projection of the characteristic directions for the ARMO group (a–e) and the TUMU group (f–h). Symbols are the same as in Fig. 5.

4.2.6. D2b Northern Late-Rhyacian metaluminous monzogranites: PESA, ROCO, MATI and ORGA groups

These granites were sampled at four localities (Fig. 1b) along the northern coast of French Guyana. Apart from group ORGA, NRM intensities are the weakest observed in this study, between 1 and 0.1 mA/m. This is probably due to a low magnetic content. Demagnetization (Fig. 9a, c, e and g) is more or less noisy but always with a similar direction (Fig. 9b, d, f and h). Compared with most of the other sites, it seems that the magnetic carrier is a mineral from the titanomagnetite family with

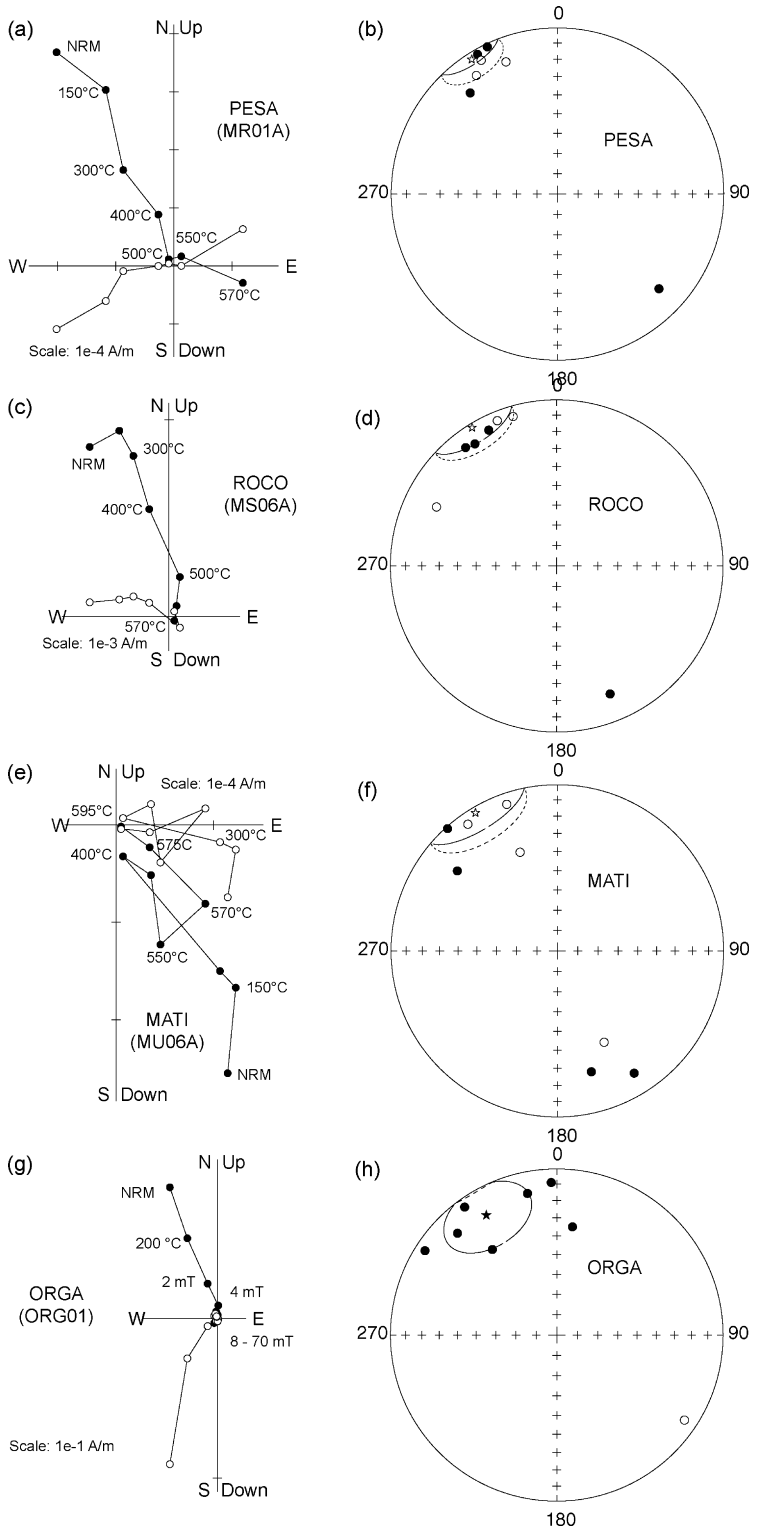


Fig. 9. Representative orthogonal vector diagrams for thermal and AF demagnetization of specimens from the late monzogranitic intrusions and equal-area projections of the characteristic group directions: (a and b) PESA-Petit Saut, (c and d) ROCO-Roche Corail, (e and f) MATI-Matiti, (g and h) ORGA-Organabo. Symbols are the same as in Fig. 5.

varying unblocking temperature but dominantly around 500 °C. The mean directions (Table 3; Fig. 9b, d, f and h) are well defined with a few reversed polarity directions. Groups ORGA, ROCO and ORGA (Table 3) were dated in the 2070–2060 Ma range. MATI group with similar paleomagnetic directions (Table 3) also belongs to the same time period (Delor et al., 2001).

4.2.7. Baked contact and reversal tests, and weathering

4.2.7.1. Baked contact test. A few backed contact tests have been presented above between Paleoproterozoic formations and younger Proterozoic formations or Early Jurassic doleritic dikes. They were performed in all studied areas except in the southernmost Tumuc Humac Mountain where no later intrusives nor dike are known. They all showed that apart from a few centimetres or up to 2 m from the intrusive rocks, a distinct paleomagnetic signal was preserved. These examples confirm that no major remagnetization occurred with the different plutonic and dike emplacements independently of their ages and with clear confidence when samples were taken away from a few meters, sometimes less, from the intrusive rocks. The main tests in Cayenne (Fig. 5c, e and g), along the Tampok (BCT, Fig. 6t and u), in the Approuague area (MATA03 group, Fig. 7f), and along the Armontabo River (ARMO group, Fig. 8e) sometimes between Proterozoic rocks give stronger evidence on the Paleoproterozoic magnetization identified during this study. Only a few groups showed Early Jurassic like directions. However, these groups are from the Tampok area (TAMP06 and TAMP08 groups) where the only dike intrusion is not Jurassic in age and with a positive baked contact test while $^{40}Ar/^{39}Ar$ biotite ages of plutonic rocks are clearly Paleoproterozoic.

4.2.7.2. Reversal test. As discussed above for the different areas and groups, most of the populations showed normal and reversed polarities. Reversal tests (McFadden and Mc Elhinny, 1990) were performed for all groups presenting both polarities. Only two groups showed a negative reversal test (MATA01 and TUMUC), the others were dominantly indeterminate (Class I) and only one of them was positive of Class C (APPR03). Although these tests may appear inconclusive it is worth noting that these reversals were observed on similar lithologies within groups. We can also suppose that this may be linked to the normal scattering of the group directions (small *k*, large a95) which resulted from slight but indeterminate local tilting which has been cancelled at the mean direction scale and probably not at the sample scale by combining multiple sampling sites within the same lithology. Therefore, presence of both polarities rather than positive reversal tests indicates that the magnetization was acquired on a sufficiently long period to record reversals and cancel any possible secular variation effect.

4.2.7.3. Weathering. It appears that the strong weathering which affects the Guiana Shield has not already remagnetized our Proterozoic samples as the main magnetic carrier appears to be magnetite which is totally chemically changed to goethite and/or hematite on weathered profiles in French Guiana (Théveniaut and Freyssinet, 1999, 2002). We only noticed, scarce and partial contribution of possible goethite with a direction similar to the Present Earth Field one as a possible effect on the low temperature component of the beginning of the present day weathering process. Some samples have a small hematite contribution with directions similar to those carried by magnetite which is never the case for weathered rocks where no magnetite exists as soon as the rock is weathered at the base of a profile (Théveniaut and Freyssinet, 1999).

4.2.8. Magnetization and geochronology

On all studied areas, we found distinct paleomagnetic directions between rocks of distinct ages in close locations. Both magnetic polarities have been identified on most formations. No pervasive remagnetization between Paleoproterozoic formations or due to the Early Jurassic volcanic activity was evidenced nor any effect of recent weathering. These indications enable us to present the different directions identified on Paleoproterozoic rocks as characteristic directions acquired during uplift and cooling for the plutonic rocks and during or at the end of the metamorphism phase for the metamorphic rocks. An important question is the age of the identified magnetization relative to the age of the studied rock formation. This problem is usually less important on Phanerozoic rocks and especially on sedimentary or volcanic rocks. However, for Proterozoic time the question is highly relevant. Indeed, assigned plutonic rock ages usually correspond to crystallisation ages of minerals like zircon or monazite which are high temperature ages compared to the age of acquisition of the remanence. It is therefore difficult to present paleomagnetic poles or apparent polar wander path with estimated magnetic ages that have no apparent equivalent to geological event. An example is the recent preliminary paleomagnetic results from French Guiana (Nomade et al., 2001, 2003). An age of 2036 ± 14 Ma is assigned to the pole OYA (Nomade et al., 2001) while poles GUY1 and GUY2 (Nomade et al., 2003) have assigned ages of

2014 ± 27 and 1993 ± 25 Ma, respectively. These ages are estimated magnetic ages based on proposed constant cooling rate at the French Guiana scale. They are also younger than all dated Paleoproterozoic formations, the youngest late metaluminous monzogranite being dated at 2060 Ma. It is therefore difficult to present the paleomagnetic results with their estimated magnetic ages without taking care of discussing the rock formation ages to allow any geologic and paleogeographic comparison with other continents. Thermal evolution and cooling history of Paleoproterozoic shields is therefore prerequisite to any discussion of paleomagnetic results obtained on Paleoproterozoic rocks. In French Guiana, for Paleoproterozoic formations, a first attempt (Nomade et al., 2002) was based on comparison of $^{40}Ar/^{39}Ar$ amphibole and biotite ages obtained from four rocks collected on a 150 km section along the Oyapock River. This led to an estimated cooling rate of 3–5 °C/Ma from 550 to below 250 °C for the 2050–1930 Ma period. A second attempt (Nomade et al., 2003) extrapolated amphibole and biotite ages, not always from the same rock, to calculate magnetic ages for two paleomagnetic poles (GUY1 and GUY2). Their magnetic ages revealed younger ages (2014 and 1993 Ma) than the OYA pole (2036 Ma) while the zircon ages (Delor et al., 2003a,b) present the opposite configuration between the Mana intrusion used for pole GUY01 (2132 Ma) and the OYA pole (2098 Ma). It is likely that using biotite ages may be locally acceptable but probably not at a scale as large as suggested by Nomade et al. (2002) from the Oyapock area. Recent $^{40}Ar/^{39}Ar$ ages (Tables 1 and 2 and Enjolvy, 2004) do not confirm the estimated cooling rate and the spatial distribution of ages suggested by Nomade et al. (2002), revealing that estimated cooling rates can strongly vary for rocks collected within a small area.

In Fig. 10, we present all available multimethod ages from French Guiana. These ages are from Delor et al. (2003a), Nomade et al. (2002) and the present study. When available, we added zircon ages on rocks which were previously dated by the $^{40}Ar/^{39}Ar$ method by Nomade et al. (2001, 2003). We estimated a mean temperature for crystallisation of gabbroic intrusions (ITA40 and GN) around 900 ± 50 °C and around 750 ± 50 °C for other acidic plutonic intrusions, which are also assigned to the crystallisation temperature of zircon and, therefore closure temperature of their U–Pb system. Closure temperature for argon is estimated at about 550 ± 30 °C for amphibole (Dahl, 1996) and 300 ± 25 °C for biotite (Harrison et al., 1985).

The distribution of U–Pb zircon and $^{40}Ar/^{39}Ar$ amphibole and biotite ages is consistent with the respective isotopic closure temperatures of these minerals. It is

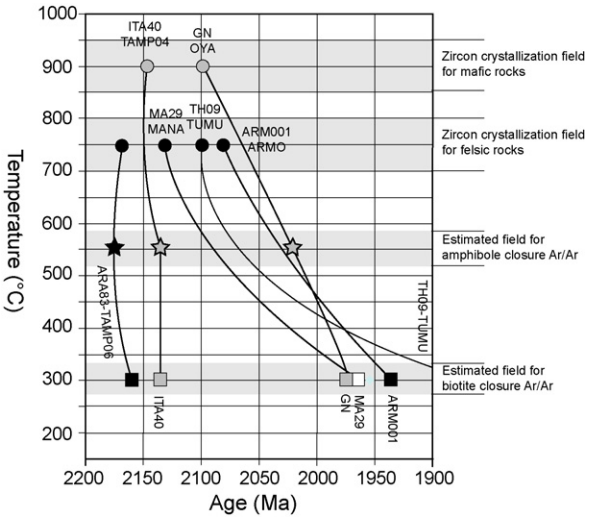


Fig. 10. Available geochronological data relative to the present paleomagnetic study with dating on zircon (U/Pb and Pb/Pb, circles), amphibole ($^{40}Ar/^{39}Ar$, stars) and biotite ($^{40}Ar/^{39}Ar$, square) with estimated cooling. Grey, black and white correspond to mafic, felsic and metamorphic, respectively.

noteworthy that the scattering of biotite ages is more pronounced than for the other minerals, probably because of its higher sensitivity to argon loss effects. Even though amphibole data are scarce, there is an apparent consistency between younging of zircon ages and younging of amphibole ages from the same samples. For biotite ages, some problem may arise as detailed above. This may reflect different cooling history and varying cooling rates from different areas or rocks. Samples from the Tampok area (ITA40 and ARA83) reveal similar ages for zircon and amphibole around 2160–2135 Ma implying an extremely fast cooling rate from 900–700 to 550 °C at this time. This cooling rate remains high until 300 °C for both samples, but the scattering of ages obtained on biotite ARA83 (Fig. 3d) (core mean around 2134 ± 18 Ma) suggests that some thermal event occurred later, giving rise to a partial isotopic resetting. On the other hand, the Mana plutonic intrusion (MA29) that metamorphosed sediments in northwestern French Guiana (Nomade et al., 2003) show different behaviour. A biotite age of 1964 ± 4 Ma was obtained (Nomade et al., 2003) from the sediments intruded by this granite. The Mana granitic intrusion has a zircon crystallisation age at 2132 ± 3 Ma (Delor et al., 2003a). As the biotite age is associated to the cooling history of the intrusive granite, and approximate cooling rate of 2–3 °C/Ma may be proposed in this area. For the GN sample of a dioritic body from the eastern Oyapock River, Nomade et al. (2001) obtained amphibole and biotite ages. A zircon age, at 2098 ± 2 Ma was also obtained from Delor et al.

(2003a). There is an apparent consistency on a nearly constant cooling rate, around 5 °C/Ma between zircon and amphibole ages and also between amphibole and biotite ones. However, these variations in cooling rates, and the use of biotite ages, have strong implications on dating of the paleomagnetic signals. For example, MA29 and GN samples which have similar biotite ages, less than 10 Ma difference, produced highly different paleomagnetic signals (Nomade et al., 2001, 2003). These differences may be more consistent with their differences in zircon ages of 34 Ma and probably of amphibole ages than extrapolated from the only available biotite ages.

From these observations, we can suggest that sequential timing of paleomagnetic poles may be associated to zircon ages of rock formations. Amphibole dating could be, when available, the best proxy for a magnetic age. However, biotite dating has to be taken cautiously to calculate a magnetic age unless it is well constrained by zircon and amphibole ages.

On the following discussion, paleomagnetic poles will be presented according to their zircon crystallisation (rock formation) ages before being linked to less available amphibole ages, This will facilitate the presentation of virtual geomagnetic poles (VGPs) from French Guiana before discussing the APWP and drift rate of the Guiana Shield.

5. Discussion

5.1. French Guiana Paleoproterozoic paleomagnetic poles

In order to compare all group directions obtained from the different areas of French Guiana, we calculated virtual geomagnetic poles (VGPs) for each group mean direction (Table 4). We also assumed that zircon dating, although different from the age of magnetic acquisition, present similar old to young sequence as discussed above. Even though zircon ages do not correspond to the age of magnetization, this allows to present paleomagnetic poles with reference to geologically assumed rock ages. The VGPs have been calculated with an arbitrary choice of the paleomagnetic polarity as already discussed (Section 4.2) in order to minimize distances between successive pole positions.

In Fig. 11a, we represented the oldest VGPs for the plutonic groups from all areas of the present study. For the Mesorhyacian rock formations, from TAMP06 at 2155 Ma to TAMP05 at 2112 Ma, the poles lie on a small circle, close to the 210°E meridian and centred on French Guiana. This may indicate some intra-cratonic rotations within French Guiana. However, if this can stand for the Tampok area, the CAYE02 VGP indicates a more realistic intra-cratonic rotation between the southwestern Tampok area and the northeastern Cayenne area. Indeed, TAMP04 and CAYE02 poles, from two similar gabbroic intrusions, are similar in age but differ significantly revealing a possible 50° clockwise rotation between them after magnetization. According to the sinistral wrenching which affected French Guiana during the Late Rhyacian D2a phase, this clockwise rotation is consistent with geological knowledge. Indeed, Northern French Guiana may have undergone some local or regional clockwise rotations relative to the southwestern similar formations both being separated by the Central Mesorhyacien TTG core.

For rocks belonging to the Neo Rhyacian D2a phase, the VGP locations extend from the granitic TAMP05 pole at 2112 Ma to the gabbroic APPR03 pole and its granitic paleomagnetically equivalent ARMO pole at 2080 Ma. During this period, there is a sequential organization with the pole moving from the initial 210°E meridian and at 15°N latitude to Mexico around 2100 Ma (TUMU pole) then changing to northeastern South America around 2095–2090 Ma and moving to the South Atlantic Ocean around 2080 Ma. The only discordant VGPs are the MATA01 and MATA02 poles which are badly defined with a95 greater than 20° (Table 3).

For younger formations of the Late Rhyacian D2b phase, the VGP locations evolve (Fig. 11b) from the South Atlantic Ocean nearly to Antarctica. The four poles from the Northern late monzogranitic intrusions of French Guiana are similar one each other and distinct from the previous poles being located close to the Antarctic continent. Assuming the opposite polarity for these youngest poles would locate their VGPs close to the Alaska, between the CAYE02 and TAMP04 poles. However, this would imply a faster and unrealistic drift rate between the ARMO-APPR03 poles at 2080 Ma and the poles around 2070–2060 Ma.

In Fig. 11c, we represented all poles obtained from the Eo to Mesorhyacian metamorphic rock formations. Two of them have been dated (CAYE01 and TAMP02) while the others are geologically linked to the Mesorhyacian D1 event which occurred between 2180 and 2130 Ma. Geological arguments and VGP locations are in close agreement with all calibrated VGPs lying close to the 210°E meridian which defined the plutonic APWP (Fig. 11a) part for rocks of this period. The only outlier pole is the MATA03 pole which is ill defined with an a95 greater than 20°. Geological arguments and geochronological results are in agreement with paleomagnetic results and associated tests. Indeed no metamorphic rock revealed a paleomagnetic signal similar

Table 4
Paleomagnetic pole positions (VGPs) derived from the data of Table 2, Nomade et al. (2001) and recalculated from Nomade et al. (2003)

Paleomagnetic group	Lithology	PGV Latitude (°N)	PGV Longitude (°E)	dp	dm
Cayenne Island					
CAYE01	Gneiss	4.6	212.6	15.4	30.5
CAYE02	Diorite/tonalite	75.0	208.6	4.2	8.4
Tampok and Ouaki rivers					
TAMP01	Metavolcanite	40.8	221.4	7.8	15.3
TAMP02	"Ouaki" orthogneiss	38.6	216.4	6.0	12.0
TAMP03	Granite	−6.9	300.1	15.9	16.1
TAMP04	Gabbro-diorite	28.0	199.7	6.3	11.8
TAMP05	Granite	14.8	211.5	10.1	20.1
TAMP06	Granodiorite	77.2	137.0	5.9	11.4
TAMP07	Gneiss-amphibolite	13.7	214.3	6.5	13.0
TAMP08	Orthogneiss	77.5	165.4	9.9	19.4
Mataroni and Approuague rivers					
MATA01	Granite	17.5	231.4	19.6	36.0
MATA02	Granite	14.9	289.2	40.6	42.7
MATA03	Metavolcanite	−14.2	257.9	22.4	31.0
APPR01	Metavolcanite	37.1	224.7	8.0	15.6
APPR02	Granite	4.5	298.9	19.1	19.2
APPR03	Gabbro	−24.8	336.6	8.4	10.1
APPR04	Metavolcanite	3.2	208.6	13.2	25.6
APPR05	Granodiorite	−5.9	296.9	34.3	35.1
APPR06	Granite	−18.5	294.3	21.3	23.0
APPR07	Granite	30.1	211.1	12.2	224.3
APPR08	Granite	5.3	293.4	16.8	17.2
Armontabo river					
ARMO	Mg–K granite	−2.7	346.3	14.2	16.9
Tumuc Humac Mountains					
TUMU	Mg–K granite	18.9	273.7	19.2	22.3
Coastal late monzogranite					
PESA	Granite	−56.7	25.1	6.2	12.4
ROCO	Granite	−58.0	26.4	7.9	15.8
MATI	Granite/Granodiorite	−58.6	25.5	9.7	19.4
ORGA	Granite	−59.7	44.7	10.1	19.5
Recalculated poles from Nomade et al. (2003)					
MANA	Mana river metasediments	62.9	235.5	55.6[a]	9.1[b]
OYA1	Oyapock TTG-granites	62.1	245.0	11.6[a]	15.8[b]
OYA2	Oyapock Granites	5.3	227.8	30.6[a]	14.0[b]

Latitude (°N); longitude (°E), dp and dm errors around the pole position.
[a] *k*-Value.
[b] a95.

to the one evidenced from the massive younger granitic intrusions which occurred around 2100–2090 Ma. Moreover, late events like the late monzogranitic intrusion only affected the northern part far from the metamorphosed studied rocks.

5.2. Guiana Shield Paleoproterozoic paleomagnetic poles

The French Guiana VGPs appear well organized according to the age of the rock formations. All poles from plutonic rocks older than 2115 Ma and for all metamorphic formations lie close to the 210° meridian while for younger plutonic formations they move from around 15°N–210°E up to the Antarctic vicinity passing through French Guiana. These poles may be used to define the French Guiana and Guiana Shield APWP. However, it is prerequisite to evaluate the available data of the Guiana Shield and to calibrate this APWP according to estimated magnetic ages.

In Fig. 12a and b, we plotted all available Paleoproterozoic paleomagnetic poles from the Guiana Shield

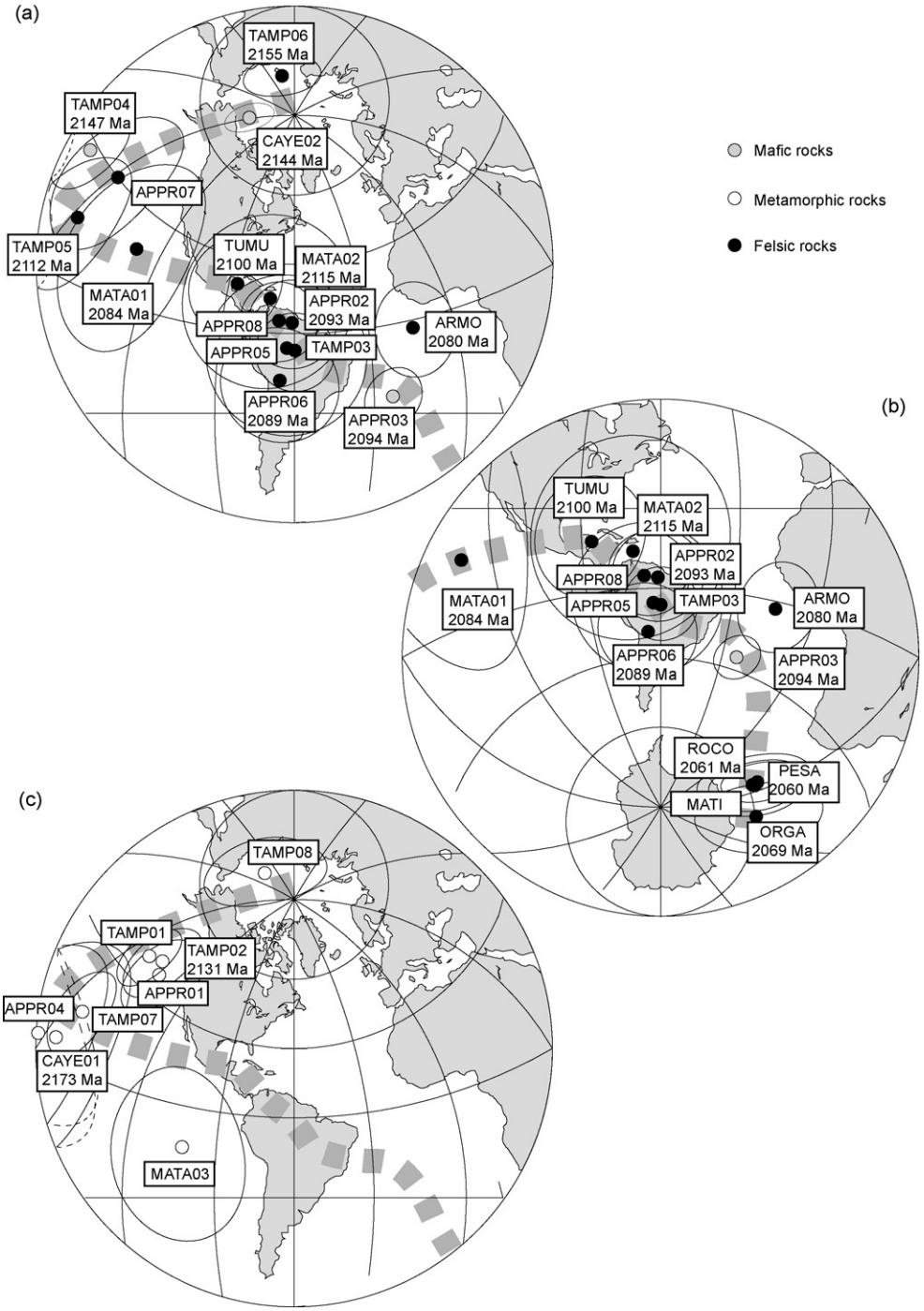


Fig. 11. (a) Stereographic projection, centred at 30°N and 300°E, of the virtual geomagnetic poles (VGPs) from French Guiana groups for plutonic rocks dated between 2155 and 2080 Ma. (b) Idem centred at 30°S and 300°E for plutonic rocks dated between 2100 and 2060 Ma. (c) Idem centred at 30°N and 300°E for metamorphic rocks. All VGPs are in Table 3.

with the French Guiana APWP defined by the successive PVG locations of Fig. 11. Some results already come from French Guiana (Nomade et al., 2001, 2003) as preliminary findings of our comprehensive sampling. They did not integrated newly obtained zircon ages nor the completely revised Formations and chronology definitions of Delor et al. (2001, 2003a). The pole OYA (Nomade et al., 2001) which comes from the Oyapock

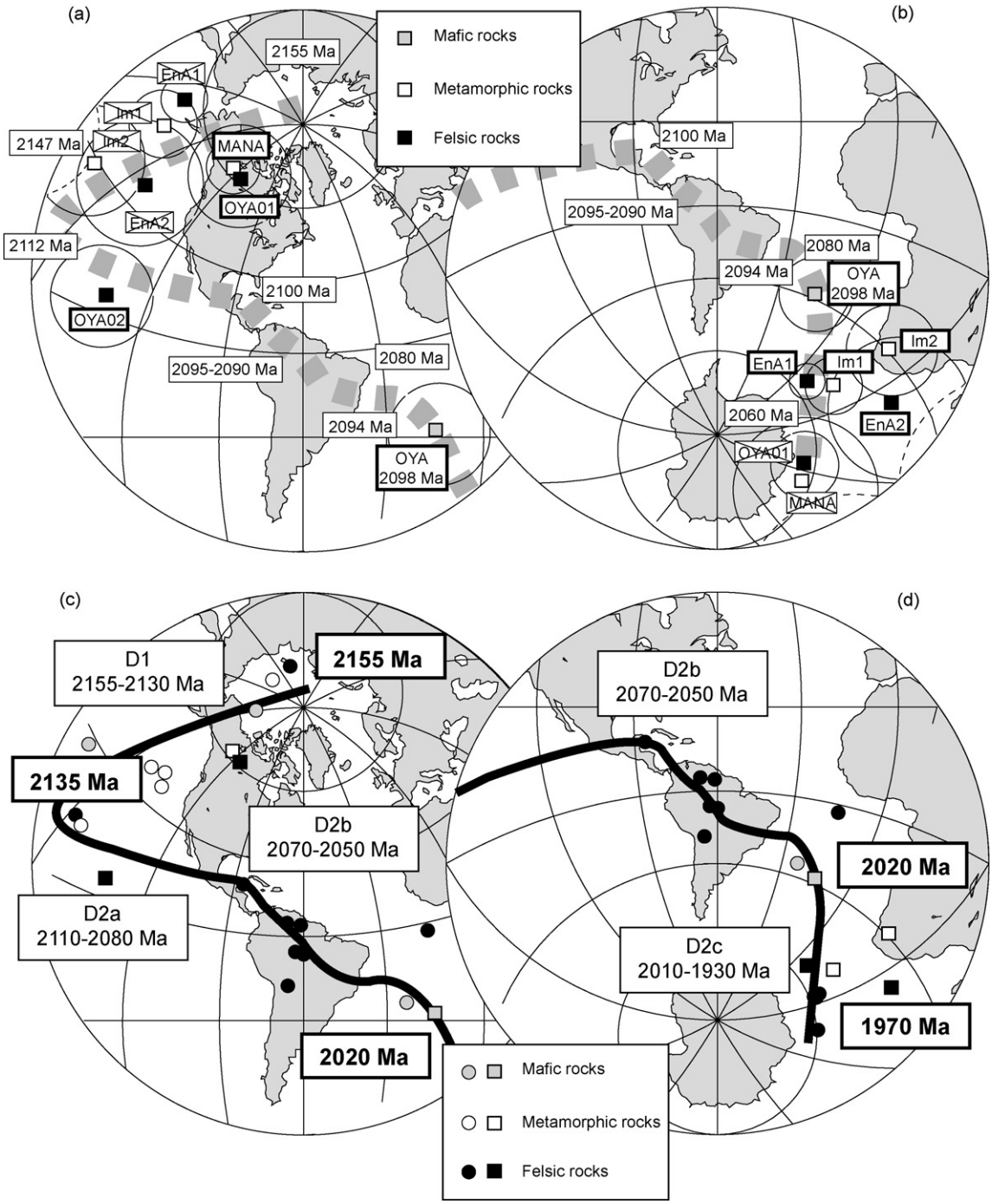


Fig. 12. (a) Available paleomagnetic poles of the Guiana Shield compared to the French Guiana APWP (stereographic projection centred at 30°N and 300°E); (b) Comparison of the same poles, but with opposite polarity (stereographic projection centred at 30°S and 300°E) (c) Guiana Shield APWP for the Mesorhyacian (D1 phase) to Orosirian (D2c phase) period (stereographic projection centred at 30°N and 300°E) and (d) for the Late Rhyacian (D2b phase) to Orosirian (D2c phase) period (stereographic projection centred at 30°S and 300°E).

River is close (Fig. 12a and b) to our APPR03 pole, similar in age and lithology. For the other poles, both possible polarities are presented in Fig. 12a or b, respectively. From the GUY1 and GUY2 poles of Nomade et al. (2003) we recalculated three poles. The MANA pole comes from northwestern metasediments of the Mana River which were intruded by dated granite as discussed above. The other poles, from the Oyapock River, have been renamed OYA1 and OYA2 as they are relevant to this river. These three recalculated poles are discussed down below and presented on Table 4.

The MANA pole falls close to the 210°E meridian of the 2155–2112 Ma part of the APWP (Fig. 12a) or slightly south and east of our youngest part (Fig. 12b). The zircon age (2132 ± 3 Ma; Delor et al., 2003a) of the granite which metamorphosed those sediments and the VGP location are in favour of a location on the 2155–2112 Ma part of the APWP. The OYA1 pole falls close to the MANA pole, both defining the GUY1 pole of Nomade et al. (2003). However, sampled rocks of the OYA1 pole come from an area where dated rocks range around 2160 Ma (Delor et al., 2003a). It seems impossible to date this pole close to the youngest event around 2060 Ma (Fig. 12d) as no age younger than 2100–2080 Ma as yet been evidenced in this area. For the OYA2 pole, it cannot be drawn in Fig. 12b as it falls east of Africa far from the APWP. For the other polarity choice (Fig. 12a), it falls close to the 2112 Ma pole from the Tampok area and on the track towards the 2100 Ma Southern Tumuc Humac (TUMU) pole. This is in agreement with the youngest granite dates along the Oyapock River around 2105 Ma (Lafon et al., 2003).

Other paleomagnetic poles come from Venezuela. Im1 (Imataca pole 1) and Im2 (Imataca pole 2) are from the Imataca Complex (Onstott and Hargraves, 1981) and poles EnA1 and EnA2 are from the Encrucijada pluton (Onstott et al., 1984). All these poles fall close to the 2155–2112 Ma part (Fig. 12a) or close to our 2060 Ma pole (. 12b) depending on the assumed polarity. The Im1 and Im2 poles are not sufficiently detailed for a critical review but we can notice their pole locations are in agreement with our 2070–2060 Ma poles of Fig. 12b for rocks subject to a peak metamorphic phase around 2.05–1.98 Ga (Tassinari et al., 2004). For the EnA1 and EnA2 poles, they were separated in two groups as a slight mean direction difference was observed between samples of normal polarity and others of reverse polarity. As the unblocking temperature was similar for the two populations, the authors (Onstott et al., 1984) proposed three possible mechanisms for this bipolarity. This was supposed to be discussed in detail in a second paper which has never been published. From our detailed observations in French Guiana, we can suggest a similar mechanism of slightly delayed cooling which allowed the recording of reversals of the earth's magnetic field. Therefore, their two poles could be combined in one pole in a probably median location that precisely falls on our APWP but around 2130 Ma (Fig. 12d) or 2060 Ma (Fig. 12a). Moreover, the EnA1 and EnA2 poles have $^{40}Ar/^{39}Ar$ dating at 1972 ± 4 Ma (amphibole) and 1883 ± 2 Ma (biotite) which are younger than the amphibole $^{40}Ar/^{39}Ar$ dating of the OYA pole (Nomade et al., 2001). We therefore prefer to represent them close to the 2070–2060 Ma (Fig. 12b) French Guiana poles.

Although the initial studies in Venezuela provided paleomagnetic poles which could not be used confidently on the basis of the standard reliability criteria of paleomagnetic data (Van der Voo and Meert, 1991), our results and Nomade et al.'s (2001 and 2003) reappraisal confirm their preliminary findings (Onstott and Hargraves, 1981; Onstott et al., 1984). Moreover the amphibole dating of the EnA1 and EnA2 poles may give key point for calibrating the Guiana Shield APWP.

5.3. Guiana Shield calibrated APWP

All the available paleomagnetic poles from the Guiana Shield have been obtained from rocks dated between 2172 and 2060 Ma. As already discussed the confident poles come from rocks belonging to the 2155–2060 Ma time range. As there are insufficient zircon-amphibole-biotite dating on the same rocks to calculate cooling rates and precise or estimate magnetization ages for all studied formation, it is better to assume that available amphibole ages may be the best proxy to calibrate a Guiana Shield APWP.

Four key points are available. The TAMP06 pole presents (Table 1) similar zircon, amphibole and biotite ages. An age of 2155 Ma may therefore be assumed for this pole. The TAMP04 pole has (Table 1) a similar zircon age at 2147 Ma but a slightly younger amphibole age which can be used as a proxy at 2135 Ma. Unfortunately, no amphibole dating could be obtained from any granite which produced zircon crystallization ages around 2100–2090 Ma. For the gabbroic intrusions (APPR03 and OYA) and the granite ARMO, the OYA (Nomade et al., 2001) amphibole age of 2020 Ma may be used. For the youngest period of the French Guiana late metaluminous intrusives and the Im1, Im2, EnA1 and EnA2 poles from Venezuela, an age of 1970 Ma may be proposed for the magnetization according to the amphibole dating (Onstott et al., 1984) of the Encrucijada pluton.

It is therefore possible to present a Guiana Shield apparent polar wander path (Fig. 12c and d) for the

geological 2155–1970 Ma time period and to evaluate the relationships between the paleomagnetic signals and the geological events which affected the entire region. The framework of the geodynamic context is discussed in the light of a recent synthesis (Delor et al., 2003b) at the Guiana Shield scale.

In Fig. 12c and d, we represent the Guiana Shield APWP with all available paleomagnetic poles discussed in Figs. 11 and 12a and b. The ages are those related to the paleomagnetic signals based on amphibole $^{40}Ar/^{39}Ar$ dating as the best proxy for paleomagnetic estimated ages. The geological events are superimposed on the track to evaluate the Guiana Shield evolution in connection with geodynamic events. During the Mesorhyacian D1 phase of subduction regime, the paleomagnetic pole moved from a location close to the present day north pole (Fig. 12c) along the 210°E meridian and to the equator at around 2130 Ma. This polar evolution characterizes only rotation of the Guiana Shield at a location close to the Equator (Fig. 13). This period reflects a convergence phase with emplacement of multi-pulse TTG suites which may have cooled fast compared to the following stages. During the Neo Rhyacian, between 2110 and 2080 Ma, granitic accretion reflects a collisional context with sinistral shearing of a D2a phase. Rock formations and slow cooling prevented any paleomagnetic record of this phase during which the pole moved along the 15° latitude toward the East. Possible intracratonic rotations may have taken place during this phase. From a paleomagnetic point of view, this sinistral deformation due to continuous deformation after a convergence is associated to a displacement (Fig. 13) of the Guiana Shield from the Equator to circum polar latitudes. Between 2070 and 2050 Ma, at a time late metaluminous monzogranitic intrusions cristallize in French Guiana, the paleomagnetic signal records evolution of the pole over the Guiana Shield from the Central America towards the South Atlantic. This Late Rhyacian D2b phase takes place, close to the pole (Fig. 13) at a time the Guiana Shield is affected by granulite-belt formation. Indeed, this period is marked by the formation of three granulite belts of Imataca, Bakhuis and eastern Amapá (Delor et al., 2003b) which mark a high geothermal heating associated to mantle upwelling. Even though no rock provided younger ages for the paleomagnetic studied formations, age estimates of paleomagnetic poles reach younger geological periods with geodynamic importance at the Guiana Shield scale. Indeed, between 2010 and 1930 Ma, granulite metamorphism (Roraima-Kanuku-Coeronie belt) is evidenced and also strong acid plutono-volcanism, pro parte "Uatumã". The Guiana Shield APWP marks an evolution of the pole location from East of South America towards the Antarctic border. During this D2c phase, the Guiana Shield moves from nearly polar latitudes towards the Equator. The D2 geological phase which extends from 2110 to 1930 Ma is marked by a slow cooling rate compared to what is evidenced from the D1 phase. This confirms the continuous magmatism and sometimes metamorphism which affect the Guiana Shield during the post-convergence D2 phase. Mantle upwelling and important heat flux stand during this period to end with a low depth heat flux as

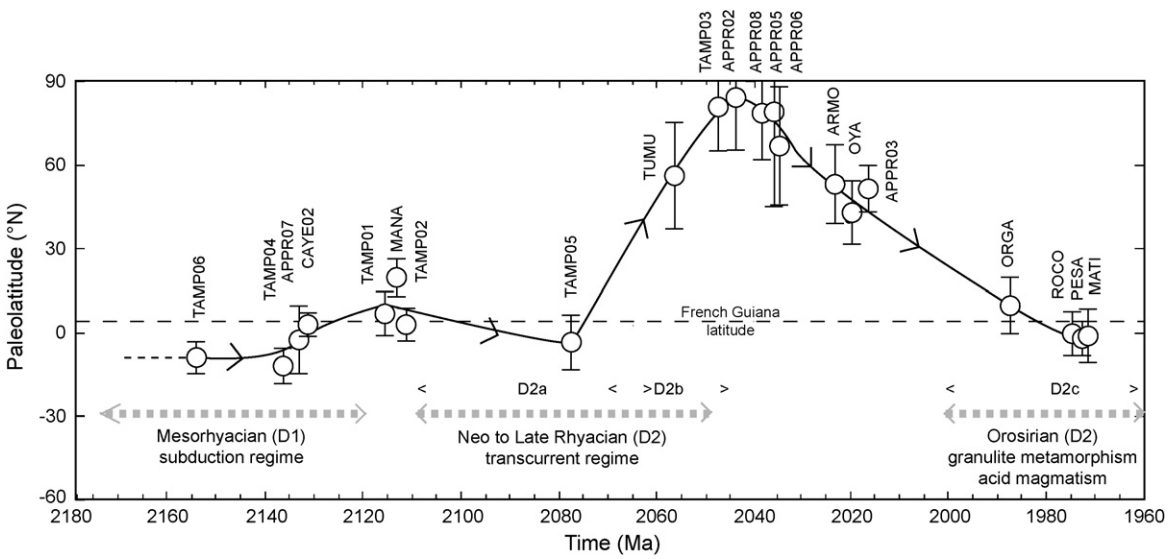


Fig. 13. Paleolatitudinal evolution through time of French Guiana between Mesorhycian at 2155 Ma and Orosirian at 1970 Ma.

evidenced by the Orosirian volcanism which marks the end of the Transamazonian event.

In terms of continental drift (Fig. 13), the best drift rate estimate can be calculated from the TAMP04 to the OYA pole for which zircon and amphibole datings are available on two similar gabbroic rocks. Taking into account the error bars in paleolatitude and ages, this gives a continental drift of 28 ± 4 to 36 ± 5 cm/y with the zircon dating while the amphibole dating gives 12 ± 3 to 16 ± 4 cm/y. The amphibole dating are supposed to be closer to the magnetic ages, as discussed earlier and can be taken as representative of the Paleoproterozoic continental drift rate for the Guiana Shield. For younger period, similar calculation can be obtained between the OYA pole and the Late Metaluminous intrusions. A continental drift estimate of 7.5 ± 1 to 18 ± 3 cm/y is derived from zircon dating while it is of 6 ± 1 to 14 ± 2 cm/y for amphibole dating. Although fast, these rates will need confirmation by new amphibole ages on other rocks and especially granitic rocks for which no amphibole dating is presently available.

Comparing paleomagnetic results, with the derived paleolatitudinal evolution, to the geological history of the Guiana Shield (Fig. 13) highlights some points as already mentioned above. During the Mesorhyacian D1 phase, between 2155 and 2130 Ma, the Guiana Shield just moves around the Equator at a time magmatic accretion dominates with volcano-sedimentary deposits subsequently metamorphosed and intruded by dominantly tonalite-trondhjemite-granodiorite (TTG) plutonism. This event is interpreted as a response to subduction of Eorhyacian oceanic crust during the convergence of West African and Amazonian Archean plates. After closure of the Eorhyacian ocean, during the Neo Rhyacian D2a phase, from 2112 to 2080 Ma, the collision is completed and transcurrent regime dominates. This is associated with the beginning of a strong paleolatitudinal evolution from the Equator towards the pole. At the time the Late Rhyacian D2b phase occurs, around 2060 Ma, the Guiana Shield moves from around 60° latitude to the pole and then back to 60° latitude but on the opposite meridian. During the Orosirian D2c phase, all the studied rocks pass through the temperature they record a paleomagnetic signal of a continent moving back to the Equator. The slow cooling of rocks of the D2 phase reflects the effect of the continuous mantle upwelling and continental heat flux which will end around 1930 Ma. This comparison of geological events and paleolatitudinal evolution may indicate strong relationship between regional tectonic regime (subduction followed by transcurrent with mantle heat flux activity) and global tectonic reorganization of continental assembly and drift.

6. Conclusion

We presented a well constrained Guiana Shield APWP for the 2155–1970 Ma period (Meso-Rhyacian to Neo and Late Rhyacian and Orosirian time) based on paleomagnetic results supported by a sufficient number of sites and samples and completed by positive field tests. Although still questionable due to possible intracratonic rotations, it can be extended to the 2173–2155 Ma period (Meso Rhyacian) on the basis of Western Guiana selected area where no Neorhyacian magmatic resetting is evidenced by remarkably concordant U/P zircon ages and $^{40}Ar/^{39}Ar$ amphibole and biotite ages. The poles are sequentially organized according to their "zircon" ages and in agreement with the presently available amphibole dating. This is the first step for establishing a Paleoroterozoic APWP which will have to be precised by new data from other parts of the Guiana Shield. On the other hand, combined geochronological dating (e.g. U/Pb or Pb/Pb on zircon and $^{40}Ar/^{39}Ar$ on biotite and/or amphibole) will be helpful in deriving a cooling rate to extrapolate a magnetic age when enough information will be available on the magnetic mineralogy (unblocking temperature range) and its relationship with the different mineralogical parageneses. We showed that intracratonic rotations probably occurred before 2112 Ma. The most important is French Guiana and probably the Guiana Shield moved from nearly the Equator before 2130 Ma to the pole around 2050 Ma before going back to the Equator at 1970 Ma on the nearly opposite meridian. These displacements have apparently strong links with geological events which marked the Transamazonian orogeny.

Acknowledgements

We are particularly indebted to the field trip organization of J.P. Richard in Cayenne, the logistical support of the French Army (9ème RIMA) for the Tumuc Humac field trip and for the help in sampling from all collaborators of the French Guiana geological mapping project (University of Orléans, University of Rennes and CPRM, Brazil). H.T. thanks G. Hugues and B. Borsier for their help in sampling preparation and analyses. Two anonymous reviewers are warmly thanked for their detailed and constructive comments.

References

Avelar, V.G. de, Lafon, J.M., Delor, C., Guerrot, C., Lahondère, D., 2003. Archean crustal remnants in the Easternmost part of the Guiana Shield: Pb–Pb and Sm–Nd geochronological evidence for

Mesoarchean versus Neoarchean signatures. Géologie de la France 2–3–4, 83–99.

Castillo, J.H., Constanzo-Alvarez, V., 1993. Paleomagnetism of the Uairen Formation, Roraima Group, southeastern Venezuela: evidence for one of the oldest (Middle Proteroizoic) depositional remanent magnetizations. Can. J. Earth Sci. 30, 2380–2388.

Dahl, P.S., 1996. The effects of composition on retentivity of Ar and O in hornblende and related amphiboles: a field tested empirical model. Geochim. Cosmochim. Acta 60, 3687–3700.

Deckart, K., Féraud, G., Bertrand, H., 1997. Age of Jurassic continental tholeiites of French Guyana, Suriname and Guinea: implications for the initial opening of the Central Atlantic Ocean. Earth Planet. Sci. Lett. 150, 205–220.

Delor, C., Lahondère, D., Egal, E., Marteau, P., 2001. Carte géologique de la France à 1/500 000, Guyane, 2ème éd., Orléans, BRGM.

Delor, C., Lahondère, D., Egal, E., Lafon, J.-M., Cocherie, A., Guerrot, C., Rossi, P., Truffert, C., Théveniaut, H., Phillips, P., Avelar, G.V., 2003a. Transamazonian crustal growth and reworking as revealed by the 1:500 000 scale geological map of French Guiana (2nd ed.). Géologie de la France 2–3–4, 5–57.

Delor, C., de Roever, E., Lafon, J.-M., Lahondère, D., Rossi, P., Cocherie, A., Guerrot, C., Potrel, A., 2003b. The Bakhuis ultrahigh-temperature granulite belt (Suriname): II. Implications for the late Transamazonian crustral stretching in a revised Guiana Shield framework. Géologie de la France 2–3–4, 207–230.

Egal, E., Milesi, J.P., Ledru, P., Cautru, J.P., Freyssinet, P., Thieblemont, D., Vernhet, Y., 1994. Ressources minérales et évolution lithostructurale de la Guyane. Carte thématique minière à 1:100 000. Feuille Cayenne. BRGM Rep. R 38 019, 59 pp.

Enjolvy, R., 2004. Caractérisation géochronologique et pétrologique de l'évènement fini-transamazonien: etude en Guyane française et en Amapá (Brésil). Rapport de DEA, Univ Montpellier II, 42 pp.

Fisher, R.A., 1953. Dispersion on a sphere. Proc. R. Soc. Lond. A 217, 295–305.

Gibbs, A.K., Barron, C.N., 1993. The Geology of the Guiana Shield. Oxford University Press, 246 pp.

Hargraves, R.B., 1968. Palaeomagnetism of the Roraima dolerites. Geophys. J. R. Astr. Soc. 16, 147–160.

Hargraves, R.B., 1978. Problems in palaeomagnetic synthesis illustrated by results from Permo-Triassic dolerites in Guiana. Phys. Earth Planet. Interiors 16, 277–284.

Harrison, T.M., Duncan, I., McDougall, I., 1985. Diffusion of ^{40}Ar in biotite: temperature, pressure and compositional effects. Geochim. Cosmochim. Acta 49, 2461–2468.

Hocquard, C., Schmitt, J.M., 1980. Carte géologique de la France à 1/100 000, Département de la Guyane, Feuille Araoua, Carte et Notice Explicative, BRGM.

Irving, E., 1956. Palaeomagnetic and palaeoclimatological aspects of polar wandering. Geofis. Pura Appl. 33, 23–41.

João, X.S.J., Marinho, P.A.C., 1982. Catametamorfitos Arqueanos da região centro-este do Território Federal do Amapá. In: Simp. Geol. Amaz. 1, Belem. Anais Belem, SBG 2, pp. 207–228.

Kirschvink, J.L., 1980. The least-square line and plane and the analysis of paleomagnetic data. Geophys. J. R. Astr. Soc. 139, 133–143.

Lafon, J.M., Delor, C., Théveniaut, H., Krymsky, R., Santos Tavares, R.P., Roig, J.Y., 2003. Isotopic deciphering of Rhyacian crustal evolution along the northern Oyapock river. New constraints from Sm–Nd, U–Pb and Pb–Pb geochronology. In: VIII Simposio de Geologia da Amazonia, Manaus, Brésil, Novembre 2003.

Lima, M.I.C., Oliveira, E.P., Tassinari, C.G.G., 1982. Cinturões granulíticos da porção septentrional do Craton Amazônico. Anais do Simposio de Geologia da Amazônia, Belém, vol. 1, pp. 147–162.

Marot, A., 1988. Notice de la carte géologique du Sud de la Guyane à 1/500 000. BRGM, 1 map, 86 pp.

Maurel, O., Monié, P., Respaut, J.P., Leyreloup, A.F., Maluski, H., 2003. Pre-metamorphic ^{40}Ar/^{39}Ar and U–Pb ages in HP metagranitoids from the Hercynian belt (France). Chem. Geol. 193, 195–214.

McElhinny, M.W., Mc Williams, M.O., 1977. Precambrian geodynamics—a palaeomagnetic view. Tectonophysics 40, 137–159.

McFadden, P.L., Mc Elhinny, M.W., 1990. Classification of the reversal test in paleomagnetism. Geophys. J. Int. 103, 725–729.

Miron-Valdespino, O.E., Constanzo-Alvarez, V., 1997. Paleomagnetic and rock magnetic evidence for inverse zoning in the Parguaza batholith (southwestern Venezuela) and its implications about tectonics of the Guiana Shield. Precambrian Res. 85, 1–25.

Montalvão, R.M.G., Tassinari, C.C.G., 1984. Geocronologia précambriana do Território Federal do Amapá (Brasil). In: Simp. Amaz., 2 Manaus. Anais Manaus, MME-DNPM, pp. 54–57.

Nomade, S., Chen, Y., Féraud, G., Pouclet, A., Théveniaut, H., 2001. First Palaeomagnetic and ^{40}Ar/^{39}Ar study of Palaeoproterozoic rocks from the French Guiana (Camopi and Oyapok rivers), northeastern Guiana Shield. Precambrian Res. 109, 239–256.

Nomade, S., Chen, Y., Pouclet, A., Féraud, G., Daouda, Y., Théveniaut, H., Rigollet, C., 2003. The Guiana and West African shields grouping. New palaeomagnetic data for the French Guiana and the Ivory-Coast. Geophys. J. Int. 154, 677–694.

Nomade, S., Féraud, G., Chen, Y., Pouclet, A., 2002. Thermal evolution of the Palaeoproterozoic Transamazonian orogen as deduced from ^{40}Ar/^{39}Ar along the Oyapok River (French Guiana). Precambrian Res. 114, 35–53.

Nomade, S., Théveniaut, H., Chen, Y., Pouclet, A., Rigollet, C., 2000. Palaeomagnetic study of French Guiana Early Jurassic dolerites: hypothesis of a multistage magmatic event. Earth Planet. Sci. Lett. 184, 155–168.

Onstott, T.C., Hargraves, R.B., 1981. Proterozoic transcurrent tectonics: palaeomagnetic evidence from Venezuela and Africa. Nature 289, 131–136.

Onstott, T.C., Hargraves, R.B., York, D., Hall, C., 1984. Constraints on the motions of South American and African Shields during the Proterozoic: I. ^{40}Ar/^{39}Ar and paleomagnetic correlations between Venezuela and Liberia. Geol. Soc. Am. Bull. 95, 1045–1054.

Pesonen, L.J., Elming, S.A., Mertanen, S., Pisarevsky, S., D'Agrella-Filho, M.S., Meert, J.G., Schmidt, P.W., Abrahamsen, N., Bylund, G., 2003. Palaeomagnetic configuration of the continents during the Proterozoic. Tectonophysics 375, 289–324.

Piper, J.D.A., 1982. The Precambrian palaeomagnetic record: the case for the Proterozoic Supercontinent. Earth Planet. Sci. Lett. 59, 61–89.

Piper, J.D.A., 1987. Paleomagnetism and the Continental Crust. Open University Press, Milton Keynes, 434 pp.

Rogers, J.J.W., Santosh, M., 2002. Configuration of Columbia, a Mesoproterozoic supercontinent. Gondwana Res. 5, 5–22.

Rosa-Costa, L.T., Ricci, P.S.F., Lafon, J.M., Vasquez, M.L., Carvalho, J.M.A., Klein, E.L., Macambira, E.M.B., 2003. Geology and geochronology of Archean and Paleoproterozoic domains of the southeastern Amapá and northwestern Pará, Brazil—southeastern Guiana Shield. Géologie de la France 2–3–4, 101–120.

Runcorn, S.K., 1956. Paleomagnetic comparisons between Europe and North America. Geol. Assoc. Can. Proc. 9, 77–85.

Santos, J.O.S., Potter, P.E., Reis, N.J., Hartmann, L.A., Fletcher, I.R., McNaughton, N.J., 2002. Age, source, and regional stratigraphy of the Roraima Supergroup and Roraima-like outliers in northern

South America based on U–Pb geochronology. Geol. Soc. Am. Bull. 115 (3), 331–348.

Tassinari, C.C.G., Munha, J.M.U., Teixeira, W., Palacios, T., Nutman, A.P., Sosa, C.S., Santos, A.P., Calado, B.O., 2004. The Imataca Complex, NW Amazonian Craton, Venezuela: crustal evolution and integration of geochronological and petrological cooling histories. Episodes 27 (1), 3–12.

Théveniaut, H., Freyssinet, P., 1999. Paleomagnetism applied to lateritic profiles to assess saprolite and duricrust formation processes: the example of the Mont Baduel profile (French Guiana). Palaeogeogr. Palaeoclimatol. Palaeoecol. 148, 209–231.

Théveniaut, H., Freyssinet, P., 2002. Timing of lateritization on the Guiana Shield: synthesis of paleomagnetic results from French Guiana and Suriname. Palaeogeogr. Palaeoclimatol. Palaeoecol. 178, 91–117.

Vanderhaeghe, O., Ledru, P., Thiéblemeont, D., Egal, E., Cocherie, A., Tegyey, M., Milesi, J.P., 1998. Contrasting mechanism of crustal growth Geodynamic evolution of the granite-greenstone belts of French Guiana. Precambrian Res. 85, 1–25.

Van der Voo, R., Meert, J.G., 1991. Late Proterozoic paleomagnetism and tectonic models: a critical appraisal. Precambrian Res. 53, 149–163.

Veldkamp, J., Mulder, F.G., Zijderveld, J.D.A., 1971. Paleomagnetism of Surinam dolerites. Phys. Earth Planet. Int. 4, 370–380.

Zijderveld, J.D.A., 1967. A.C. demagnetization of rocks: analysis of results. In: Collinson, D.W., Creer, K.M., Runcorn, S.K. (Eds.), Methods in Paleomagnetism. Elsevier, Amsterdam, pp. 254–286.

Weil, A.B., Van der Voo, R., Mac Niocaill, C., Meert, J.G., 1998. The Proterozoic supercontinent Rodinia: paleomagnetically derived reconstructions for the 1100 to 800 Ma. Earth Planet. Sci. Lett. 154, 13–24.

Zhao, G., Cawood, P.A., Wilde, S.A., Sun, M., 2002. Review of global 2.1–1.8 Ga orogens: implications for a pre-Rodinia supercontinent. Earth Sci. Rev. 59, 125–162.